Cloudra: Geometric Reasoning and Scientific Validation for Autonomous Reaction Mechanism Exploration

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Abstract

Autonomous exploration of organic reaction mechanisms requires molecular geometries, reaction pathways, and quantum-chemical results to support the same mechanistic conclusion. Large language model (LLM) agents can make unreliable three-dimensional judgments and declare a study complete before the required physical checks are satisfied.

We present Cloudra, a multi-agent framework that maintains explicit correspondences among chemical intent, molecular identity, geometric operations, and quantum-chemical evidence. Under a design principle of intelligent restraint, topology-aware deterministic operators translate chemical intent into spatial operations on molecular structures. A Delivery DAG organized around chemical entities links mechanistic claims to archived quantum-chemical results, checks stationary points and reaction pathways, and keeps unresolved scientific requirements active through geometric operations, verification, and completion decisions.

We evaluate Cloudra on CloudraBench, which contains 50 tasks covering elementary reactions, multistep mechanisms, stereochemical competition, and computational method limitations. Across 200 runs under four workflow configurations, Cloudra achieves a mean composite score of 0.793, compared with 0.581 for the same-model Standard baseline. On Hard-tier tasks, the mean score increases from 0.689 to 0.958. The self-consistency component reaches its maximum score in 24 of the 40 non-Impossible tasks, compared with seven under Standard. Behavioral analyses examine how explicit geometric measurements and structured result associations affect local scientific judgments.

Cloudra couples computational execution to geometric operations and physical acceptance checks. The framework evaluates completion through the results supporting a mechanistic conclusion and the scientific requirements that those results satisfy.

1 Introduction

1.1 Physical requirements for autonomous reaction mechanism research

Scientific validation of an organic reaction mechanism requires checking the molecular geometries, reaction pathways, and quantum-chemical results assigned to each species and elementary step. Normal termination of a geometry optimization (exit 0) does not establish that the result is a stable intermediate, and one imaginary frequency does not by itself identify the saddle point of the intended reaction. An intrinsic reaction coordinate (IRC) that reaches an unintended dissociation product or stereoisomer also changes the mechanistic assignment of the associated energy calculation. These checks are particularly important when a study includes several elementary steps or competing stereochemical pathways.[1, 2]

Computational chemistry agents automate several parts of this work. El Agente Q [3], Aitomia [4], and Gráfico [5] orchestrate quantum-chemical calculations and typed workflows. Estructural [6] and ARCHE [7] provide tools for molecular fragment editing, atom mapping, and transition-state geometry preparation; Quntur [8] addresses reaction kinetics and transition-state research. Autonomous mechanism exploration also requires reliable geometric operations and a completion decision based on scientific validation.

Aligning stereochemical orientations, assembling reactant fragments, and preparing pathway endpoints require three-dimensional geometric reasoning. Language models can make errors when they infer spatial relationships or directly manipulate Cartesian coordinates. Quntur [8], for example, reports the need for human assistance with endpoint alignment in complex pathway and transition-state preparation. This motivates interfaces through which a model can specify a chemical objective and inspect the resulting geometric change.

Completion requires checking whether the computed species and steps satisfy the physical acceptance conditions. Intermediates require confirmation of zero imaginary frequencies, while the frequency and bidirectional IRC results for a saddle point must correspond to the same reaction. An agent can otherwise declare a mechanism complete while these checks remain unresolved. A system for autonomous exploration therefore needs to retain the chemist's research objective, the acceptance requirements, and the computational artifacts used to evaluate them.

1.2 Cloudra: a physical contract-driven system for autonomous mechanism exploration

We introduce Cloudra, a multi-agent system for exploring reaction mechanisms involving small organic molecules. Language models propose mechanistic hypotheses, choose exploration pathways, and assess computational setbacks. Deterministic software maintains atom-topology correspondences and performs local geometric operations. Physical acceptance checks evaluate whether the returned structures, pathways, and mechanistic conclusions meet the task's scientific requirements.

Figure 1 summarizes a structured literature sample of 85 chemical AI agent systems: 39 concern materials or surfaces, 23 concern finite molecules or complexes, 12 concern soft matter or biomolecules, and 11 cover multiple categories of objects. Figures 2 and 3 show the documented tasks in the finite-molecule/complex category and compare Cloudra's capabilities with those of the existing systems. The accompanying literature materials contain the complete matrix and the sources for each entry.

Cloudra uses the principle of intelligent restraint to assign

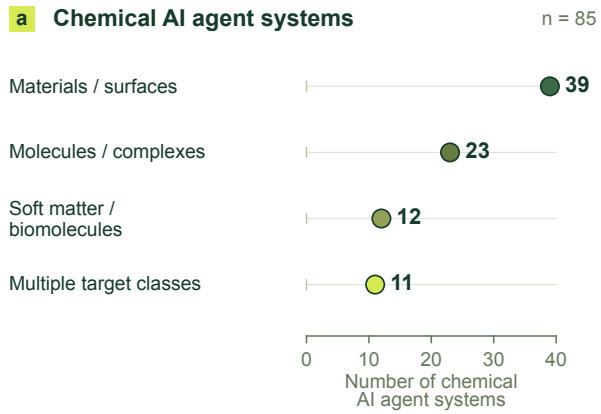


Figure 1. Physical targets in a structured literature sample of 85 chemical AI agent systems: 39 materials/surface, 23 finite-molecule/complex, 12 soft-matter/biomolecular, and 11 cross-class systems.

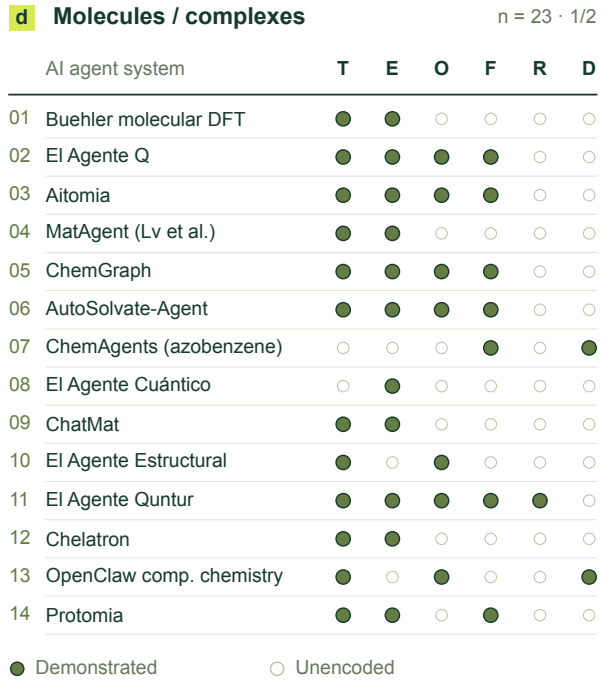


Figure 2. Documented tasks in the finite-molecule/complex class (panel 1 of 2; 23 chemical AI agent systems in total). T: structure construction/editing; E: static energy/properties; O: geometry optimization; F: vibrations/thermochemistry; R: explicit pathways/TS; D: molecular/quantum dynamics. Filled cells denote positive evidence; blanks are unencoded, not absent capabilities.

geometric calculations and coordinate changes to deterministic operators. A maximum common-subgraph (MCS) algorithm retrieves molecular fragments. Atom mapping and reaction-center identification connect molecular topology to the coordinates used in subsequent operations. Measurement operators return bond lengths, bond angles, dihedral angles, and stereochemical features; manipulation operators perform fragment rotation, translation, and transition-state assembly. The model specifies the intended structural transformation, examines the measured change, and selects the next configuration. This interface reduces the need for manual endpoint alignment.

Cloudra organizes scientific validation through a directed acyclic graph centered on chemical entities, the Delivery

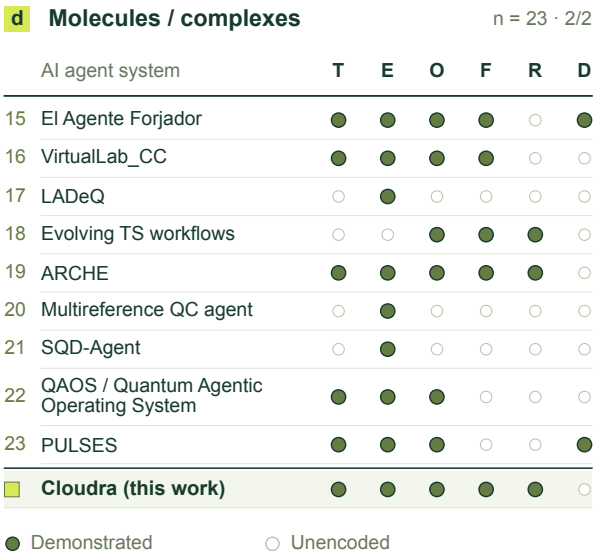


Figure 3. Chemical AI agent systems in the finite-molecule/complex class, continued (panel 2 of 2), using the codes in Figure 2. Cloudra (this work) is benchmarked at the bottom. Complete matrices for all 85 chemical AI agent systems and source-level qualifications accompany the draft.

DAG. Ground-state chemical species and elementary reaction steps provide the references for the associated computational artifacts. Each step carries requirements for a first-order saddle point with exactly one valid imaginary frequency and for bidirectional IRC evidence. The graph binds each claim to the original quantum-chemical logs used in its evaluation. The Supervisor reviews the results against the task's scientific contracts alongside the programmatic acceptance checks. A step with unresolved frequency or pathway requirements remains unconfirmed when the model requests completion.

1.3 Contributions

Cloudra addresses a common requirement of autonomous mechanism research: chemical intent, molecular identity, geometric operations, and computational evidence must continue to refer to the same species and elementary steps. The Delivery DAG assigns scientific validation requirements to these objects, checks the underlying quantum-chemical artifacts, and records unresolved requirements so that completion requests can be evaluated against physical acceptance conditions throughout an extended research trajectory.

Deterministic procedures connect molecular identity and atom correspondence to geometric operations and target-specific measurements. These procedures include MCS-based topological mapping, steric graph traversal, and transition-state fragment assembly. Language models use them to investigate chemical hypotheses, compare competing pathways, and revise unsuccessful calculations during autonomous transition-state searches.

We evaluate these choices with CloudraBench, which measures graph coverage, physical self-consistency, and task-defined target accuracy from native computational outputs. Its 50 tasks cover elementary and multistep reactions, stereochemistry, and the applicability limits of the allowed methods. The experiments comprise 48 independent runs with three repeats on four tasks and a completed study of 200

runs under four workflow configurations across all 50 tasks. Standard uses the same model backbone as the primary comparison; Record and Brief-errors examine evidence organization and diagnostic feedback.

2 System Design and Methods

2.1 System Overview and Division of Responsibilities

Cloudra takes reactant identities, reaction conditions, target species or mechanistic questions, and computational requirements as inputs. It represents candidate elementary steps in a delivery evidence graph, generates molecular configurations, submits quantum chemical calculations, and associates returned structures, energies, frequencies, and paths with the corresponding species or steps. The graph stores proposed and verified chemical objects together with unresolved calculation requirements.

The multi-agent system divides responsibilities by functional role. The Coordinator formulates mechanistic hypotheses, proposes candidate elementary reaction steps, and maintains exploration branches. Computational agents execute assigned reaction steps, invoking geometric operators to assemble three-dimensional coordinates and scheduling quantum chemical calculations. When an assigned step encounters geometric divergence, saddle-point nonconvergence, or pathway bifurcation, the computational agent assesses whether the step remains viable; if the current step appears obstructed, it feeds diagnostic information back to the Coordinator to adjust the hypothesis or switch branches. The Supervisor independently audits the delivery graph against task specifications, maintaining review notes separately from parsed computational facts.

2.1.1 Asynchronous Computation Scheduling and Resource Management

The computational execution harness is governed by a central calculation ledger (`ComputeDB`), which enforces state management across the lifecycle of each job. Dedicated capacity pools partition execution across local execution threads, remote SSH clusters, and Slurm supercomputing queues, enforcing hardware concurrency quotas to prevent queue overflow. Prior to dispatch, the scheduler computes an immutable SHA-256 content digest from Cartesian coordinates, electronic spin and charge, and level of theory; identical active or cached jobs are reused to achieve deduplication. Process liveness and execution concurrency are tracked via operating-system file exclusive locks (`flock`), avoiding termination of background quantum-chemistry jobs holding active locks during monitor exceptions. Computational agents interact with the scheduler via event-driven notifications: upon calculation termination, the harness dispatches completion events, allowing agents to read artifacts asynchronously without polling loops.

2.2 Molecular Representations and Geometric Operators

A molecular input consists of an attributed graph $H = (A, B)$ and Cartesian coordinates $\mathbf{X} \in \mathbb{R}^{n\times 3}$. Here A is the atom set, B is the bond set, and each row of $\mathbf{X}$ gives one atom position. Atom and bond attributes encode chemical identity; coordinate operators update $\mathbf{X}$ and return measured changes. Operators that replace atoms or functional groups also update H.

2.2.1 Atom Correspondence and Reaction Centers

Atom attributes retain element, isotope, formal charge, hydrogen count, radical-electron count, and declared stereochemistry; bond attributes retain order and stereochemical labels. Components lacking three-dimensional coordinates are embedded using ETKDGv3 [9–11], then assembled if necessary. Existing coordinates are retained. Complete, unique atom-map labels on both endpoints define the correspondence directly and must agree in element and isotope. Otherwise, let A_{P} and A_{R} be the product and reactant heavy-atom sets. Candidate bijections $\pi : A_{\mathrm{P}} \to A_{\mathrm{R}}$ satisfy $z_i^{\mathrm{P}} = z_{\pi(i)}^{\mathrm{R}}$ and $\iota_i^{\mathrm{P}} = \iota_{\pi(i)}^{\mathrm{R}}$, where z is atomic number and ι is isotope label. Each candidate is scored by

$$J(\pi) = 2(N_{\mathrm{f}} + N_{\mathrm{b}}) + N_{\mathrm{o}} + 4\Delta_H + 8\Delta_q + 8\Delta_r + 2N_{\mathrm{s}}. \tag{2.1}$$

Let B_{P}^{π} denote product bonds expressed in reactant atom indices. Bond formation and cleavage are the set differences $B_{\mathrm{P}}^{\pi} \setminus B_{\mathrm{R}}$ and $B_{\mathrm{R}} \setminus B_{\mathrm{P}}^{\pi}$, with cardinalities N_{f} and N_{b}. The count N_{o} includes pairs in $B_{\mathrm{P}}^{\pi} \cap B_{\mathrm{R}}$ whose bond orders differ. For $a \in \{H, q, r\}$, the mismatch is $\Delta_a = \sum_{i\in A_{\mathrm{P}}} |a_i^{\mathrm{P}} - a_{\pi(i)}^{\mathrm{R}}|$, where H, q, and r are the atom hydrogen count, formal charge, and radical-electron count. N_{s} counts stereochemical mismatches. Exactly one declared tetrahedral label contributes one mismatch, increasing J by 2. When both corresponding atoms carry tetrahedral labels, the comparison requires four corresponding neighbors, including a single hydrogen where applicable; it corrects for neighbor-order parity before comparing the labels. Noncomparable two-sided tetrahedral environments contribute no mismatch.

For element–isotope classes of sizes n_g, the search space contains $N_\Pi = \prod_g n_g!$ bijections. All are enumerated when $N_\Pi \le 10^5$. Above this limit, MCS seeds use element matching, unrestricted bond-order matching, ring-to-ring matching, and complete rings [11]. The MCS timeout is 10 s, and each endpoint contributes at most 64 substructure matches. Unmatched atoms are paired within element–isotope classes after sorting by hydrogen count, degree, and input index. The search then merges candidates having identical bond-edit facts and score, keeping the lexicographically smallest mapping for each group. Representatives are sorted by score and then atom-map sequence; the first 16 form the heavy-atom candidate set C. For nonempty C, define optimal score $J_{\min} = \min_{\pi\in C} J(\pi)$ and tolerance set $\mathcal{N} = \{\pi \in C : J(\pi) \le J_{\min} + 2\}$. An unlabeled mapping is reported as unique when $|\mathcal{N}| = 1$, ambiguous when $|\mathcal{N}| > 1$, and inconclusive when $C = \emptyset$.

The retained heavy-atom candidates are combined with compatible bindings to both concrete endpoint graphs and expanded to all-atom correspondences. Compatible explicit hydrogen labels are preserved first; remaining local hydrogens are paired by isotope on corresponding heavy atoms. For unmatched isotope groups of sizes h_g, the remaining $N_H = \prod_g h_g!$ permutations are enumerated when $N_H < 257$. An expansion with $N_H \ge 257$ yields an inconclusive result unless compatible explicit hydrogen labels resolve the permutation space. Complete maps are deduplicated, ranked by score and atom-map sequence, and evaluated with the same +2 ambiguity criterion. The selected all-atom bijection reorders the product into reactant atom order. Formed

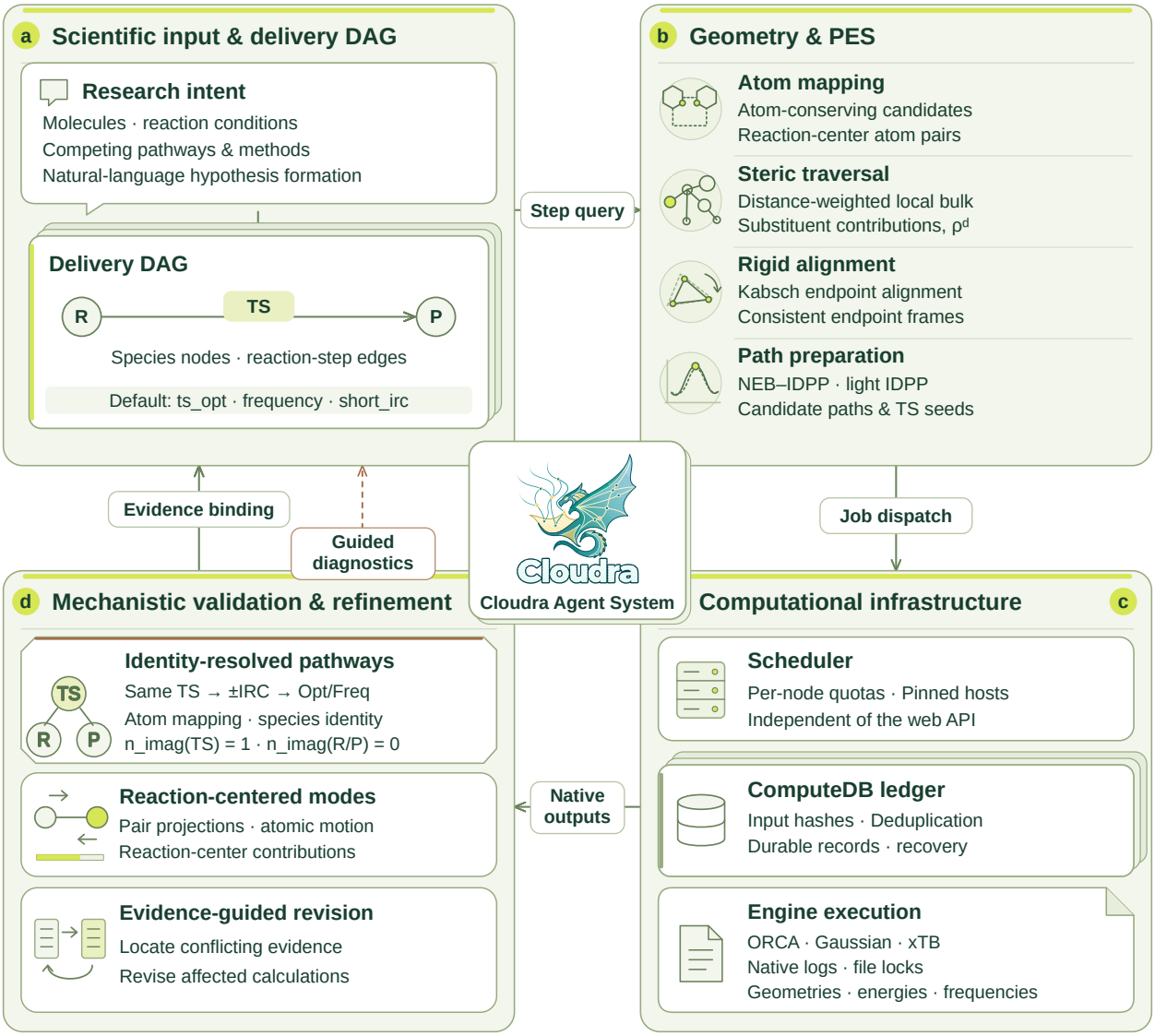


Figure 4. Cloudra multi-agent system architecture: coupling mechanistic hypothesis generation (Coordinator), deterministic geometric manipulation (Operators), asynchronous computational execution (ComputeDB), and physical evidence validation gates.

and broken bonds define the reaction-center coordinate pairs used to monitor subsequent path interpolation, while bond-order modifications are recorded alongside the transformed graph.

2.2.2 Structural Measurements and Rigid Body Operations

Structural perception converts coordinates into bond lengths, bond angles, dihedral angles, ring conformations, and close contacts. For coordinates $\mathbf{x}_i$, the interatomic distance is $d_{ij} = \|\mathbf{x}_j - \mathbf{x}_i\|$; the bond angle at vertex j is obtained from the normalized dot product of the two bond vectors. Comparisons of distances with reference bond lengths and van der Waals contact scales provide geometric diagnostics, while the molecular graph retains the established identity. Connectivity is perceived at entry for coordinate-only inputs; subsequent local coordinate transformations retain this graph and remeasure anomalies before and after transformation.

Default fragment selection on either side of an acyclic bond uses graph connectivity: the specified bond $a{-}b$ is temporarily removed, and the connected component containing b, denoted $\mathcal{F}_b$, is selected. If the endpoints remain connected after bond removal, movement within a ring framework requires an explicit atom group. Let $\mathcal{F}$ denote the explicitly specified moving atom group or the component $\mathcal{F}_b$ obtained by bond removal. To rotate the selected fragment about an axis, Rodrigues' formula is applied to every $i \in \mathcal{F}$:

$$\begin{aligned} \mathbf{x}_i' &= \mathbf{a} + \mathbf{R}(\hat{\mathbf{u}}, \theta)(\mathbf{x}_i - \mathbf{a}), \\ \mathbf{R} &= \cos\theta\, \mathbf{I} + (1 - \cos\theta)\hat{\mathbf{u}}\hat{\mathbf{u}}^{\mathsf{T}} \\ &\quad + \sin\theta\, [\hat{\mathbf{u}}]_{\times}. \end{aligned} \tag{2.2}$$

Here, $\mathbf{a}$ is a point on the axis, $\hat{\mathbf{u}}$ its unit direction, θ the rotation angle, $\mathbf{I}$ the identity matrix, and $[\hat{\mathbf{u}}]_{\times}\mathbf{v} = \hat{\mathbf{u}} \times \mathbf{v}$. Coordinates outside the selected set are unchanged. Stretching bond $a - b$ translates $\mathcal{F}_b$ as a whole:

$$\mathbf{x}_i' = \mathbf{x}_i + (L - d_{ab})\frac{\mathbf{x}_b - \mathbf{x}_a}{d_{ab}}, \qquad i \in \mathcal{F}_b. \tag{2.3}$$

Here L is the target bond length; coordinates outside $\mathcal{F}_b$ remain fixed. Chirality inversion rotates two separable branches around their center to exchange branch directions. Group substitution removes the specified-side frag-

ment, docks the new group along the original attachment-bond direction, and updates the graph. To compare structures before and after optimization, the system differences distance-perceived connectivity sets under the same element sequence and classifies gained or lost contacts as intra- or interfragment changes.

2.2.3 Branch Crowding and Reactant Preorganization

For candidate axes that pass checks of hybridization, ring structure, and substituent graph inequivalence, the system uses a decaying branch descriptor to provide local steric crowding information:

$$B_s = \sum_{k \in T_s} c_k r_k \rho^{\ell_k}, \qquad \rho = 0.5, \quad 0 \le \ell_k \le 5. \tag{2.4}$$

Here r_k represents the van der Waals radius of atom k. T_s is generated by a last-in-first-out depth-first traversal starting from the substituent root at depth zero, with the adjacent axis-end atom marked visited. On popping an unvisited atom, the algorithm records its traversal depth ℓ_k and pushes its unvisited neighbors if $\ell_k < 5$. The weighting coefficient is $c_k = 1.5$ for a root of graph degree at least 3, and $c_k = 1$ otherwise.

The descriptor uses standard van der Waals radii in Å: H 1.20, B 1.92, C 1.70, N 1.55, O 1.52, F 1.47, P 1.80, S 1.80, Cl 1.75, Br 1.85, and I 1.98 (1.70 fallback for unlisted elements). The four branch values are rounded to three decimal places and summed to obtain the axis score B_{axis}; $B_{\text{axis}} \ge 9.0$ flags the axis as congested for subsequent conformational inspection.

During multicomponent preorganization, the primary fragment is fixed at the origin. Remaining fragments undergo orientation sampling on the unit sphere. With K samples, direction s is

$$\begin{aligned} \hat{\mathbf{u}}_s &= (a_s \cos s\gamma,\ y_s,\ a_s \sin s\gamma), \\ a_s &= \sqrt{1 - y_s^2}, \qquad y_s = 1 - \frac{2s}{K-1}, \\ \gamma &= \pi(3 - \sqrt{5}). \end{aligned} \tag{2.5}$$

Here $\gamma = \pi(3 - \sqrt{5}) \approx 2.3999$ rad (137.5°) represents the Golden Angle, generating a Fibonacci spherical spiral grid for uniform distribution of directions. The axial twist about $\hat{\mathbf{u}}_s$ is $2\pi s/K$ (default $K = 32$). Translation along the interfragment vector resolves steric overlap. For a set W of guided atom pairs with target distance windows $[l_{ij}, u_{ij}]$, placement $\mathbf{X}$ is scored by

$$\begin{aligned} S(\mathbf{X}) = \max_{(i,j)\in W} \{0,\ & l_{ij} - d_{ij}(\mathbf{X}), \\ & d_{ij}(\mathbf{X}) - u_{ij}\}. \end{aligned} \tag{2.6}$$

Candidate configurations with low $S(\mathbf{X})$ are deployed as seeds for distance-restrained MMFF94 force-field optimization[12], guiding fragments toward target interaction intervals while relaxing nonbonded contacts. Qualified candidates must simultaneously satisfy force-field convergence, distance compliance, stereochemical retention, and fragment contact connectivity $d_{i^*j^*} \le 1.7(r_{i^*} + r_{j^*})$. Qualified candidates are ranked by unconstrained single-point energy and clustered with a heavy-atom Kabsch RMSD threshold of 0.5 Å to yield preorganization seeds.

2.2.4 Endpoint Alignment and IDPP Interpolation

After all-atom correspondence is established, each endpoint is centered. Let $\mathbf{P}$ and $\mathbf{R}$ be the centered product and reactant coordinate matrices. Kabsch alignment[13] solves

$$\begin{aligned} \min_{\mathbf{Q}^\mathsf{T}\mathbf{Q}=\mathbf{I}, \det \mathbf{Q}=1} & \|\mathbf{P}\mathbf{Q} - \mathbf{R}\|_\mathrm{F}^2, \\ \mathbf{P}^\mathsf{T}\mathbf{R} &= \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\mathsf{T}, \\ \mathbf{Q} &= \mathbf{U}\,\mathrm{diag}\big(1, 1, \det(\mathbf{U}\mathbf{V}^\mathsf{T})\big)\mathbf{V}^\mathsf{T}. \end{aligned} \tag{2.7}$$

The resulting matrix $\mathbf{Q}$ is the proper rigid-body rotation matrix. The determinant correction factor $\det(\mathbf{U}\mathbf{V}^\mathsf{T})$ enforces $\det \mathbf{Q} = +1$, ruling out improper reflections and preserving chiral configuration.

For M images including endpoints, define $\lambda_k = k/(M-1)$ and initialize $\mathbf{X}_k = (1-\lambda_k)\mathbf{R} + \lambda_k \mathbf{P}\mathbf{Q}$. The image-dependent pair potential (IDPP) method [14] interpolates in internal pair-distance space ($d^*_{ij,k} = (1-\lambda_k)d^\mathrm{R}_{ij} + \lambda_k d^\mathrm{P}_{ij}$), defining its potential objective as

$$\begin{aligned} d^*_{ij,k} &= (1-\lambda_k)d^\mathrm{R}_{ij} + \lambda_k d^\mathrm{P}_{ij}, \\ V_k &= \sum_{i<j} \frac{(d_{ij,k} - d^*_{ij,k})^2}{d^4_{ij,k}}. \end{aligned} \tag{2.8}$$

The inverse-fourth-power weighting $d^{-4}_{ij,k}$ imparts stiffness to short covalent bonds, prioritizing the preservation of basic covalent distances while allowing long-range nonbonded contacts to relax. Standard mode couples intermediate images with improved-tangent nudged elastic band (NEB) springs and optimizes with MDMin, evaluating convergence by the maximum force residual across movable atoms.[15, 16] Lightweight mode evaluates target distances with an objective denominator $\max(d^*_{ij,k}, 0.1\,\text{Å})^4$ and relaxes intermediate images independently using Adam optimization.[17]

Active monitoring pairs W incorporate bond-state changes, pairs bonded at both endpoints with $|\Delta d| \ge 0.15$ Å, and the five atom pairs exhibiting the largest endpoint distance variations. Within the monitoring set W, the primary reaction coordinate is selected by $p^* \in \mathrm{argmax}_{p\in W} |d^\mathrm{P}_p - d^\mathrm{R}_p|$, setting $m = (d^\mathrm{R}_{p^*} + d^\mathrm{P}_{p^*})/2$. The initial transition-state guess is extracted from the first image k satisfying $(d_{p^*,k} - m)(d^\mathrm{R}_{p^*} - m) \le 0$, or the image minimizing $|d_{p^*,k} - m|$. The selected configuration serves as the initial seed for saddle-point optimization.

2.3 Evidence Graph and Scientific Validation

2.3.1 Species Occurrences and Evidence Requirements

The delivery evidence graph is $G = (V, E)$, where vertices $v \in V$ represent chemical species occurrences and directed edges $e = (v_i, v_j) \in E$ represent elementary reaction steps. A mapping $\kappa : V \to \mathcal{S}$ maps each occurrence to its canonical chemical identity, allowing cyclical pathways to represent regenerated catalyst occurrences as distinct topological nodes sharing the same chemical identity. Stationary-point geometries, frequencies, and absolute energies attach to species nodes; transition-state geometries, imaginary vibration vectors, and reaction paths attach to elementary-step edges.

Each graph element carries a declarative stage requirement set K. Standard elementary steps require the sequence

$K = \{\text{ts_opt}, \text{frequency}, \text{short_irc}\}$, supplemented by thermochemical corrections when mandated by the task rubric. Calculation outputs are archived as content-digested immutable records. Parsed facts take the form $f = (c, o, r, d)$, where c is the check item, $o \in \{+, -, ?\}$ denotes support, contradiction, or uncertainty, r is the artifact provenance reference, and d encapsulates quantitative metrics.

For tasks declaring explicit property checks R and optional checks O, the scientific evidence acceptance criterion over target-associated facts F_t evaluates as

$$\begin{aligned} S_t = \mathbf{1}[R \neq \emptyset] \\ \wedge \bigwedge_{c \in R} \mathbf{1}[\exists f \in F_t : f.c = c,\ f.o = +] \\ \wedge \mathbf{1}[\nexists f \in F_t : f.c \in R \cup O,\ f.o = -]. \end{aligned} \tag{2.9}$$

When $R \neq \emptyset$, any missing required fact or active contradiction in $R \cup O$ invalidates S_t. When $R = \emptyset$, fact-level evaluation is omitted, delegating verification to the stage-chain requirements.

2.3.2 Curvature and Reaction Mode Diagnostics

Stationary-point curvature is evaluated from the mass-weighted Hessian:

$$\begin{aligned} \mathbf{H}_{\text{mw}} &= \mathbf{M}^{-1/2} \nabla^2_{\mathbf{x}} U\, \mathbf{M}^{-1/2}, \\ \mathbf{H}_{\text{mw}} \mathbf{v}_k &= \lambda_k \mathbf{v}_k. \end{aligned} \tag{2.10}$$

In the vibrational subspace orthogonal to translational and rotational degrees of freedom, negative eigenvalues $\lambda_k < 0$ correspond to imaginary harmonic frequencies $\nu_k < 0$. The number of imaginary frequencies is $n_{\text{imag}} = \sum_k \mathbf{1}[\nu_k < 0]$. Ground-state reaction intermediates require $n_{\text{imag}} = 0$, whereas transition states require strictly $n_{\text{imag}} = 1$.

Provenance verification confirms that the geometry evaluated by vibrational frequency analysis matches the optimized stationary point. For coordinate matrices $\mathbf{X}$ and $\mathbf{Y}$ centered at the origin, the geometry match requires $\delta_\pi(\mathbf{X}, \mathbf{Y}) = \max_i \|\mathbf{Y}_{\pi(i)} \mathbf{Q}_\pi - \mathbf{X}_i\| \leq 3 \times 10^{-6}$ Å, evaluated across element-conserving permutation mappings π and optimal Kabsch rotations $\mathbf{Q}_\pi$.

For a first-order saddle point with imaginary normal-mode displacement vector $\mathbf{u}$, the mode-alignment diagnostic compares the largest relative-displacement magnitude among monitored reaction-center pairs W with the largest among all atom pairs:

$$\eta = \frac{\max_{(i,j) \in W} \|\mathbf{u}_i - \mathbf{u}_j\|}{\max_{i<j} \|\mathbf{u}_i - \mathbf{u}_j\|}. \tag{2.11}$$

The diagnostic $\eta \in [0, 1]$ measures reaction-center participation relative to the largest pair displacement, using a comparison ratio of 0.5. It can identify spectator motion such as methyl wagging or solvent libration. The resulting alignment flag is advisory; scientific acceptance uses the first-order-saddle requirement and the same-TS bidirectional IRC and endpoint optimization/frequency chain.

2.3.3 IRC Direction and Endpoint Consistency

Intrinsic reaction coordinate (IRC) trajectories evolve from the saddle point along the steepest-descent path in mass-weighted coordinates $\mathbf{q} = \mathbf{M}^{1/2}\mathbf{x}$ [1, 2]:

$$\begin{aligned} \frac{\mathrm{d}\mathbf{q}_\pm}{\mathrm{d}s} &= -\frac{\nabla_{\mathbf{q}} U}{\|\nabla_{\mathbf{q}} U\|}, \\ \mathbf{q}_{\text{init},\pm} &\approx \mathbf{q}_{\text{TS}} \pm \epsilon \mathbf{v}_-. \end{aligned} \tag{2.12}$$

Cloudra validates that the IRC path originates at the verified saddle point and extracts the terminal frame in each integration direction. For complete-minimum pathways, dual-direction continuity is established by taking each terminal IRC frame, performing unconstrained geometry optimization, and verifying that the resulting stationary point has zero imaginary frequencies ($n_{\text{imag}} = 0$) and matches the declared chemical identity of the corresponding endpoint species in molecular topology and stereochemistry (enforced via canonical isomeric SMILES equality).

For elementary steps yielding dissociated molecular components, each dissociated product must independently converge to a verified minimum. Directional alignment is validated through the reactant-side distance interval predicate. Over the monitored pair set W, let $\mathbf{q}_{\text{TS}}$, $\mathbf{q}_{\text{E}}$, and $\mathbf{q}_{\text{R}}$ denote distance vectors at the transition state, terminal IRC frame, and reactant reference. With covalent cutoffs $c_{ij} = 1.25(r_i + r_j)$ and tolerance $\tau = 0.10$ Å, define $W_{\text{c}} = \{(i,j) \in W : d^{\text{R}}_{ij} \leq c_{ij}\}$ and interval $[a_p, b_p] = [\min(d^{\text{TS}}_p, d^{\text{R}}_p) - \tau, \max(d^{\text{TS}}_p, d^{\text{R}}_p) + \tau]$. The Boolean predicate is_between is satisfied when $\|\mathbf{q}_{\text{E}} - \mathbf{q}_{\text{R}}\|_2 < \|\mathbf{q}_{\text{TS}} - \mathbf{q}_{\text{R}}\|_2$ and $a_p \leq d^{\text{E}}_p \leq b_p$ for all $p \in W_{\text{c}}$. Dissociated product pathways require satisfaction of is_between alongside verified component minima to certify reaction continuity.

2.3.4 Evidence Aggregation and Energies

For elementary step e, let K_e denote the set of mandatory scientific stages. The stage verification predicate $P_{e,k} = 1$ indicates that stage k possesses bound computational evidence satisfying its physical acceptance criteria ($P_{e,k} = 0$ otherwise). Stage-chain completion is defined by

$$A_e = \bigwedge_{k \in K_e} P_{e,k}. \tag{2.13}$$

When explicit property checks R are declared, full step qualification requires the conjunction $A_e \wedge S_e$. Stage states are tracked through explicit dispositions (supported, pending, skipped, or rejected), distinguishing documented exemptions or limitations from completed calculation artifacts.

The electronic activation barrier is computed directly from verified stationary-point energies:

$$\Delta E^\ddagger_e = E_{\text{TS},e} - E_{\text{ref},e}. \tag{2.14}$$

Energies share a consistent quantum chemical level of theory. For thermochemical profiles, zero-point energy, enthalpy, and Gibbs free-energy corrections are evaluated from harmonic partition functions under declared temperature and pressure specifications.

2.4 Diagnostics and Recovery

The harness categorizes computational setbacks into execution errors, numerical nonconvergence, and physical evidence defects. Execution errors encompass input syntax, environmental faults, or parsing anomalies. Numerical status tracks self-consistent field (SCF) and geometry optimization convergence. Physical evidence checks identify curvature

violations ($n_{\text{imag}} \neq 1$), unverified IRC endpoints, or chemical graph discrepancies. Imaginary-mode participation and geometric interval checks provide advisory diagnostics for deciding what to inspect or revise.

Let T denote the set of declared target species and elementary steps. Extending $P_{e,k}$ across all targets $t \in T$, the outstanding scientific evidence requirements form the unsatisfied contract set:

$$U = \{(t,k) : t \in T,\, k \in K_t,\, P_{t,k} = 0\}. \tag{2.15}$$

The set U records the agent's outstanding targets. Structured diagnostic feedback maps non-zero entries in U to underlying SCF nonconvergence, normal-mode misalignment, broken IRC lineages, or chemical graph discrepancies. When a calculation fails, the harness preserves historical artifacts and updates U, providing the agent with geometric and calculation status details for adjustment.

Recovery preserves historical attempts and raw calculation outputs while rescheduling dependent stages. Atom index reordering, evidence binding, and graph status updates are handled by the harness, allowing the model to focus on modifying chemical constraints, adjusting dihedral angles, or selecting alternative electronic-structure approximations.

2.5 Implementation Defaults

The implementation described here corresponds to Cloudra revision 70daa5a.[18] Numerical arrays and linear algebra use NumPy[19]; quantum-chemical quantities are extracted with cclib alongside engine-specific native-log readers[20]; and benchmark graph representation and matching use NetworkX[21]. Preorganization uses 32 rigid orientations, up to 8 guided placements, minimum interfragment separation 1.0 Å, fragment-contact multiplier 1.7, and heavy-atom RMSD clustering threshold 0.5 Å. MMFF94 relaxation uses up to 2000 iterations per phase with a distance restraint force constant of 100.0. Standard IDPP employs 10 images optimized via MDMin (100 steps, force tolerance $f_{\text{max}} = 0.1$, NEB spring constant $k = 0.1$); lightweight IDPP uses Adam optimization (4000 steps, step size 0.02, gradient tolerance 10^{-7}). Path convergence criteria evaluate the maximum force residual on intermediate path images.

The mapping coefficients and the $J_{\text{min}} + 2$ ambiguity window are fixed ranking heuristics. Charge and radical mismatches carry larger costs than hydrogen-count and bond-edit mismatches. Branch crowding uses the fixed decay $\rho = 0.5$, traversal through depth five, and congestion flag $B_{\text{axis}} \geq 9.0$. The mode-participation diagnostic uses the comparison ratio 0.5 as an advisory flag. These values rank or annotate candidates; stationary-point and pathway evidence determine scientific acceptance.

Cloudra also provides a conformer-search tool with RDKit and external CREST backends.[22] The default preorganization described above uses rigid placements, guided MMFF94 relaxation, and clustering. CREST can be requested for additional conformer or noncovalent-complex sampling, followed by quantum-chemical optimization and physical validation.

2.6 CloudraBench construction and experimental design

2.6.1 Task coverage and tier rationale

CloudraBench contains 50 tasks: 20 Easy tasks examine elementary transformations, conformational changes, and basic execution reliability; 10 Normal tasks examine intermediate identification and multistep continuity; 10 Hard tasks examine competing pathways and stereochemical identity; and 10 Impossible tasks examine the limits of the allowed methods and justified negative conclusions (Figure 5). The larger Easy set broadens basic chemical coverage, while each of the other categories retains a distinct test. Chemical families and research requirements intersect, so related reactions can span tiers through different pathway, configuration, or electronic-state requirements. Selection considers both discriminative value and computational tractability, using literature support and reference calculations rather than assigning difficulty from reaction names alone. An accompanying task catalogue records the name, input conditions, and allowed methods for every task.

2.6.2 Reference answers and evaluation protocol

Each task fixes the chemical conditions, charge, spin, deliverables, and allowed methods. Easy, Normal, and Hard tasks permit xTB[23], PM7[24], and B3LYP-D3(BJ)/def2-SVP[25–30]; Impossible tasks are restricted to xTB and PM7. Computational evidence is read from native ORCA, Gaussian, or MOPAC logs.[31–34] Reference answers specify species, elementary steps, stereochemical identities, and applicable numerical or negative targets. Numerical barriers are electronic energy differences at a consistent level of theory; stable-minimum references and actual IRC connectivity are verified separately. For accessible reaction steps requiring verification, TS frequencies, bidirectional IRCs, outerframe provenance, and endpoint optimization/frequency calculations form the evidence chain. Species-property and electronic-state questions are checked against their specified targets.

2.6.3 Scoring and study configurations

Graph structure, self-consistency, and accuracy are evaluated separately and combined with weights of 0.1, 0.3, and 0.6. Scientific acceptance and delivery completeness are reported separately. Correct recognition of a method limitation does not require an invented barrier. Reference-evidence completeness, special treatment, and unresolved qualifications are documented per task; the presence of a reference graph and complete scientific acceptance are assessed separately.

2.6.4 Repeated-task study, primary comparison, and workflow ablations

The repeated-task study (P1b) compares Guided, Standard, Record, and Brief-errors on E1, NB1, HA1, and IA1, with three independent runs per combination (48 task-level runs). The repeated runs used DeepSeek-v4.1-flash (1017).[37] E1, NB1, HA1, and IA1 concern elimination preorganization, multistep decarboxylation, stereochemical ring opening, and electronic-structure method limits, respectively; their archived trajectories supply the subsequent local diagnostic cases. Its elapsed-time unit is a four-task batch at concurrency two, with three batches per configuration. The full-

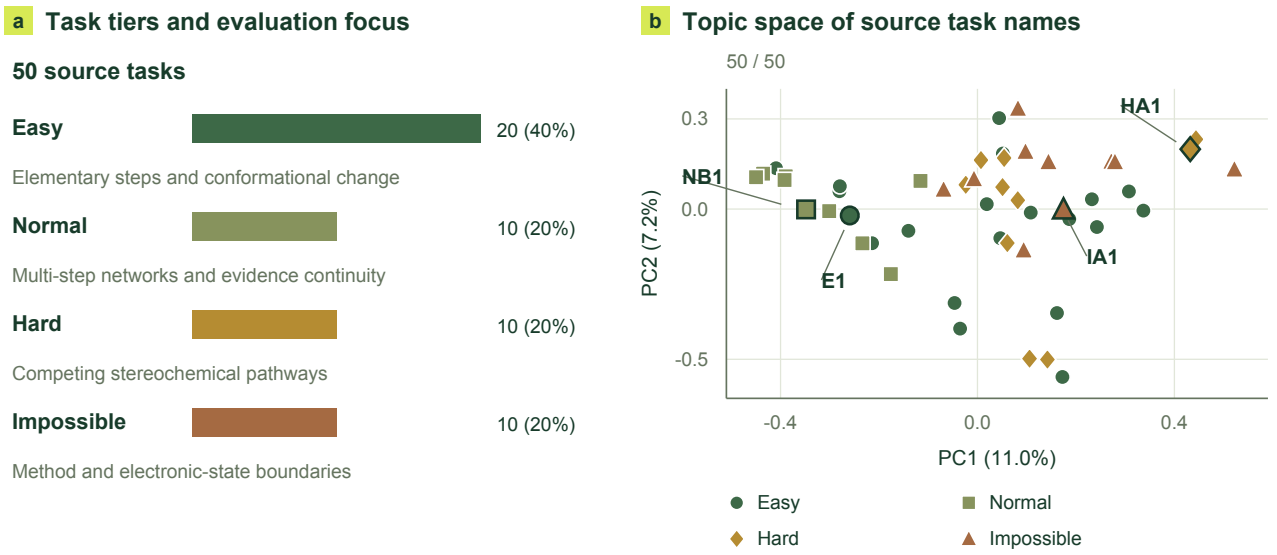


Figure 5. CloudraBench task composition and research themes. (a) Counts and evaluation focus of the four task tiers. (b) Two-dimensional PCA, computed with scikit-learn[35], of BAAI/bge-m3 embeddings[36] of all 50 original task names. E1, NB1, HA1, and IA1 are the four tasks used in the repeated-run experiment. Colors and marker shapes encode the predefined task tiers.

benchmark study (P2) retains one run for each of the 50 tasks and four configurations (200 task-level runs). All P2 configurations used DeepSeek-v4.1-flash (1017), called with the archived API alias `deepseek-v4-flash` through the official DeepSeek endpoint and streaming OpenAI-compatible chat completions, with requested minimal reasoning effort. Frozen configurations and actual request/completion records agree on this model identifier. The primary comparison is Guided against Standard using this same backbone; Standard is labelled Backbone in the figures. P1b and P2 are the archived identifiers of the repeated-task and full-benchmark studies. The P2 execution versions and request settings are listed in Supplementary Information Section S1.

Standard hides explicit DAG tools, removes scientific acceptance interception, and supplies neutral computational feedback. Guided retains the research graph, scientific checks, and detailed diagnostics. Record preserves available tools and records while removing mandatory acceptance; Brief-errors retains guidance while shortening diagnostics. Common task statements and method restrictions apply across configurations; run versions, budgets, stopping reasons, and artifact provenance are retained in the archive.

Evaluation uses mechanism graphs assembled offline from each run's conversations, tool records, reports, and computational artifacts. Each graph is linked to the run's existing score components, with field provenance retained; organization adds no conclusions from Gold. The final dataset contains all 200 runs and reports Graph, Self-consistency, and Accuracy alongside each composite score. Paired differences and 95% percentile intervals use 100,000 task-cluster bootstrap samples, retaining the four configurations of each task together (seed 20261031). Computational effort uses the 15 selected tasks shared by all four configurations in the deep-dive archive, reporting quantum-chemical job counts and cumulative job-hours. These effort measurements and the pilot's batch elapsed times are reported as separate quantities.

3 Results and Discussion

3.1 End-to-end benchmark evaluation

We evaluate Cloudra's ability to construct mechanisms, assemble the physical evidence supporting its claims, and reach task-consistent positive or negative conclusions. We report composite scores together with their graph, self-consistency, and accuracy components. Offline organization assembles each mechanism graph from its run's own research materials and links it to the run's existing score records. The resulting 200-run table contains all three components for every run, with an overall composite mean of 0.7051. Figures 6–10 distinguish final delivery, the support established for individual calculations, and agreement with the task's answer.

3.1.1 Cloudra improves research delivery relative to the same backbone

Guided obtained a mean composite score of 0.7930, compared with 0.5814 for Standard. Across the 50 task pairs, the mean improvement was 0.2116, with a 95% task-bootstrap interval of [0.1290, 0.2985]. Guided scored higher on 36 tasks, lower on seven, and equally on seven; the median paired difference was 0.1142 (Figure 6). The improvement therefore appears across many research questions rather than arising solely from a small increase in the highest-scoring runs.

Guided's mean over its ten lowest-scoring tasks was 0.3221, compared with 0.0930 for Standard.

The most consistent difference occurred in the Hard stereochemistry tasks. Guided averaged 0.9577 and Standard 0.6888, a paired increase of 0.2689 [0.1213, 0.4459], with Guided scoring higher on all ten tasks. These tasks require the system to maintain the identity of competing configurations or channels, compare the relevant barriers, and attach evidence to the corresponding conclusion. The observed improvement thus concerns the completion of a comparative chemical study, including the organization of the evidence supporting its stereochemical outcome.

On Normal tasks, Guided averaged 0.7370 versus 0.5731,

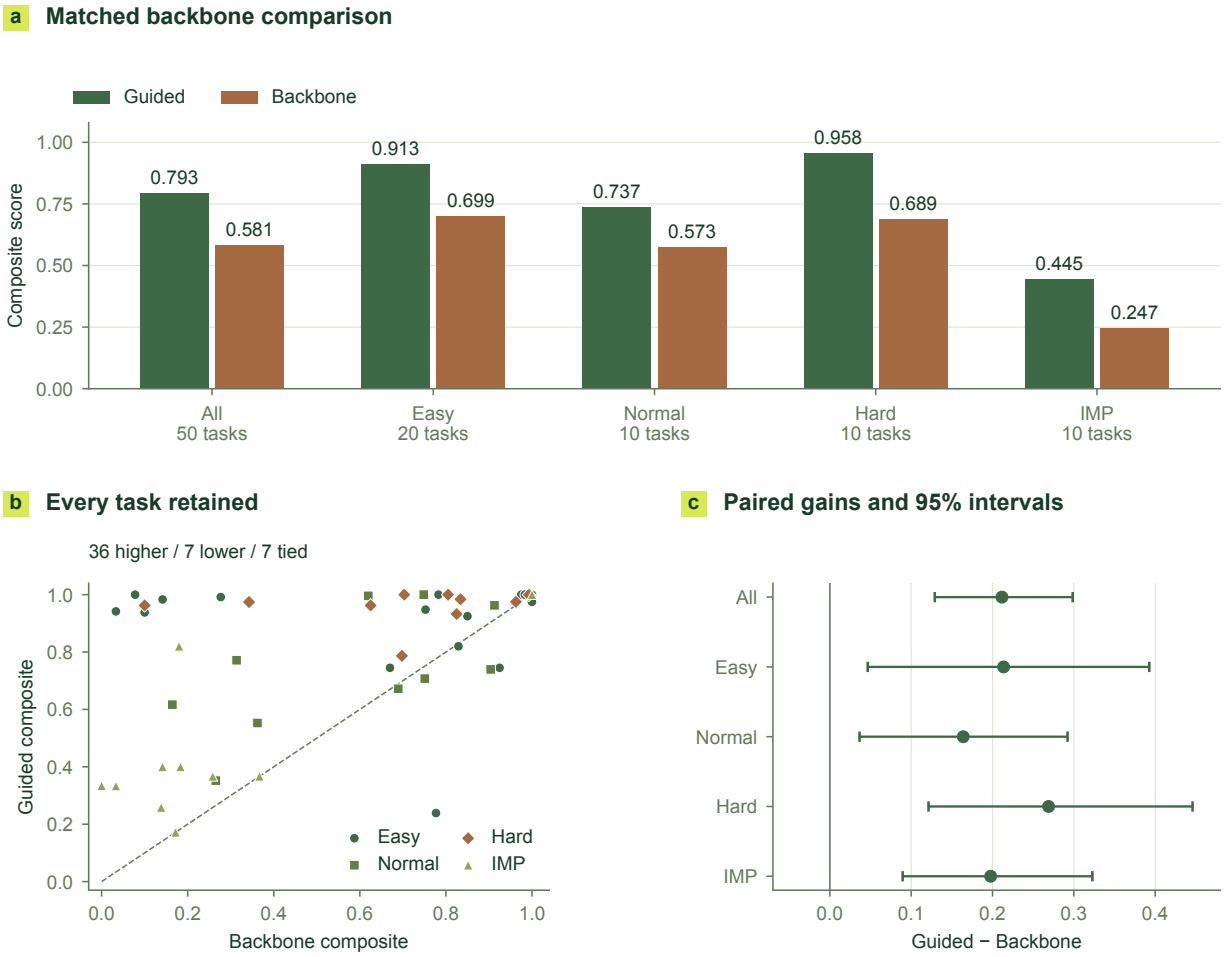


Figure 6. P2 benchmark outcomes under the archive-adjudicated track. (a) Guided and the same-backbone Standard baseline across all 50 tasks and each tier. (b) All 50 paired task outcomes. (c) Mean paired score differences with 95% percentile intervals from 100,000 task-level bootstrap resamples. Each task has one run per configuration; Easy contains 20 tasks and the other tiers contain 10 each.

an increase of 0.1640 [0.0365, 0.2922], with seven wins and three losses. Easy-task means were 0.9125 and 0.6989, respectively; the paired median difference was 0.0500, showing a different pattern from the broadly positive Hard comparison. Impossible-task means were 0.4451 and 0.2473. Their interpretation depends on whether the run addressed the requested method-limited negative target, so these tasks are examined separately below.

3.1.2 Evidence closure accounts for much of the component-level gain

Separating the three score components locates where the workflows diverged (Figure 7). Graph measures the delivered coverage and identities of the requested species and steps. Self-consistency measures the support linking saddle points, frequencies, IRC starts, and endpoint optimization/frequency calculations. Accuracy measures agreement with the task's numerical or negative target. A run can therefore give a close numerical answer while leaving its supporting path incomplete, or establish a coherent local calculation that does not answer the requested research question.

Across all 50 Guided–Standard task pairs, the weighted graph contribution to the mean improvement was +0.0108, self-consistency contributed +0.1179, and accuracy +0.0829, accounting together for the composite gain of 0.2116. For the 20 Normal and Hard tasks, the mean gain was 0.2164, with contributions of +0.0080 from graph, +0.1247 from self-consistency, and +0.0837 from accuracy. Within the ten Normal tasks, the corresponding contributions were +0.0098, +0.1141, and +0.0400; within the ten Hard tasks they were +0.0063, +0.1353, and +0.1274.

The full-task decomposition therefore locates gains in both scientific evidence closure and the requested answers. On Hard tasks, the accuracy improvement is comparable in size to the self-consistency improvement.

HH1 provides a direct example. Guided and Standard both reported a binaphthyl racemization barrier of 23.48 kcal/mol, close to the 23.01 kcal/mol reference. Both had graph scores of 1 and accuracy scores of approximately 0.973. Their self-consistency scores were 1.0 and 0.5, producing composite scores of 0.9838 and 0.8338, respectively. The archived Standard study contained candidate channels and IRC inputs, but its accepted evidence did not complete the required endpoint connections. Guided attached the path evidence to the corresponding saddle and endpoints. HI1 shows a related pattern: accuracy was approximately 1 in both runs, while self-consistency increased from 0.2333 to 1.0 and the composite score from 0.7032 to 0.9999. These comparisons identify an added layer of research completion after a promising numerical answer has already been obtained.

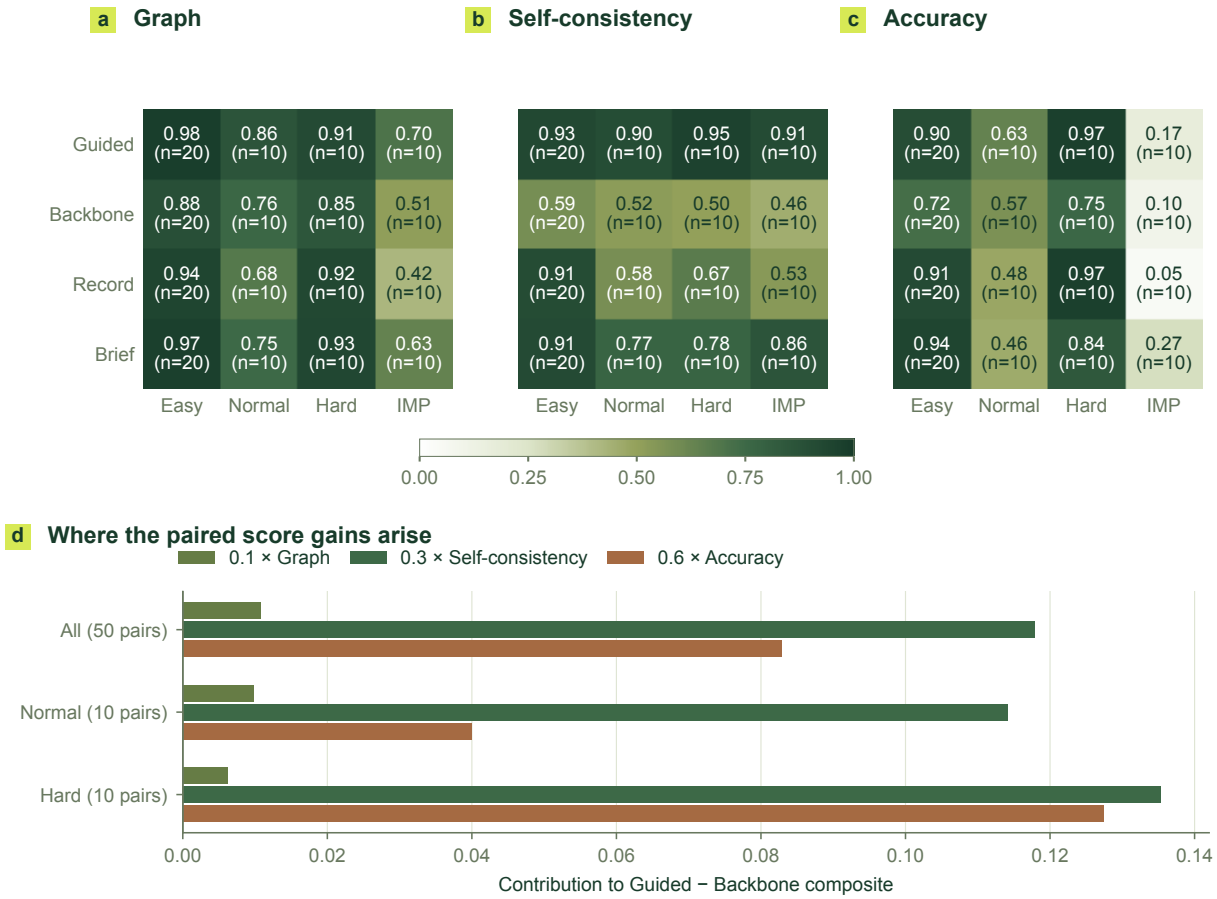


Figure 7. CloudraBench score layers across all 200 organized runs. (a–c) Means of Graph, Self-consistency, and Accuracy in each arm and tier, using 50 tasks per arm overall, 20 Easy tasks, and ten tasks in each other tier. (d) Guided–Backbone mean differences decomposed with the fixed weights 0.1/0.3/0.6, using all 50 pairs, ten Normal pairs, and ten Hard pairs. Components are aligned with the final organized totals using each run's recorded research materials and score records.

3.1.3 Delivery artifacts and graded scientific support remain distinct

The root delivery graph and the evidence carried by that graph describe different parts of the research process (Figure 8). During the original runs, Standard delivered root graphs for 43 of 50 tasks, while Guided, Record, and Brief-errors each delivered 50. The seven Standard tasks without runtime root graphs were F1, HA1, HF1, IA9, M17, NR1, and P3. Offline organization assembled their graphs from the corresponding runs' existing materials, producing a complete 50-task set for every configuration.

The seven Standard studies averaged 0.1537 after organization, compared with 0.8285 for Guided on the same tasks, and contributed 44.7% of the total Guided–Standard difference. Their individual scores ranged from 0.0333 to 0.3425. The contrast connects the runtime completion decision to the amount of research ultimately available for delivery: computations and candidate descriptions could remain in a run's records while the requested pathway or comparative conclusion had not yet been completed. The organized graph preserves that research state for evaluation.

The graded self-consistency distribution reveals a further difference among the completed task records. For the 40 non-Impossible tasks in each configuration, Guided had self-consistency exactly equal to 1 on 24 tasks and partial positive credit on 16. Standard had seven full-score, 29 partial-score, and four zero-score tasks. Record had 16 full-score and 24 partial-score tasks; Brief-errors had 20 of each. Figure 8 presents these complete evidence-score distributions alongside the separate runtime graph-delivery counts. This view preserves the amount of scientific support assembled within each study.

3.1.4 Ablations separate available capabilities from persistent guidance

Record averaged 0.6918 and Brief-errors 0.7541 across all 50 tasks. Guided exceeded Record by 0.1012 [0.0396, 0.1695], with 28 wins, seven losses, and 15 ties. On all 20 Normal and Hard tasks together, the mean Guided–Record difference was 0.1468: graph contributed +0.0083, self-consistency +0.0919, and accuracy +0.0466. Keeping the tools and a place to record results therefore produced a different outcome from keeping unresolved scientific requirements active throughout the study.

3.1.5 Elapsed research time and computational effort describe different gains

Figure 9 presents the timing information from the repeated-task study alongside the computational effort available for P2. For the repeated study, the measurement unit is a complete four-task batch at a concurrency of two, with three observed batches per configuration. Mean batch elapsed times were 4.451, 3.958, 3.894, and 4.827 hours for Guided,

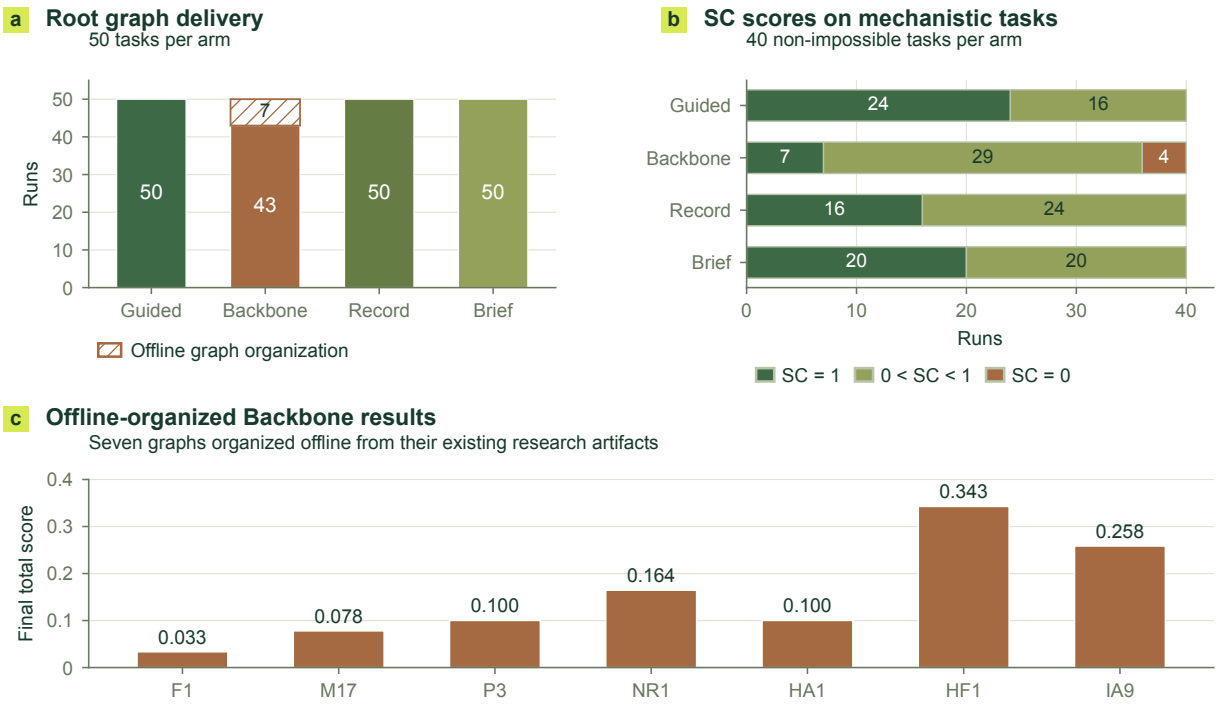


Figure 8. Runtime graph delivery and final scientific support in P2. (a) Original root mechanism graphs across 50 tasks per arm: Guided 50, Backbone 43, Record 50, and Brief 50. The hatched segment marks seven additional Backbone graphs assembled offline from the runs' own research materials; the organized set contains 50 graphs per arm. (b) Final continuous self-consistency (SC) scores for all 40 non-Impossible tasks per arm, separating SC = 1, partial positive credit, and SC = 0. Full-score counts are 24, 7, 16, and 20; Backbone has four zero-score tasks. (c) Organized composite scores for the seven Backbone tasks without runtime root graphs, ranging from 0.0333 to 0.3425 (mean 0.1537).

Standard, Record, and Brief-errors, respectively. Guided batches took 12.5% longer than Standard and 7.8% less time than Brief-errors on average. The figure shows all 12 observed batches, retaining their variation and original outcomes. These measurements include the batch's research lifecycle, waiting for calculations, and continuation after results return.

P2 effort is summarized over the 15 deeply inspected tasks shared by all four configurations. Guided used 11.08 job-hours per task, Standard 7.55, Record 7.88, and Brief-errors 15.59. The corresponding final composite means were 0.8124, 0.6495, 0.7360, and 0.7273. Guided used 46.7% more cumulative job time than Standard while increasing the mean score by 0.1629; relative to Brief-errors, it used 28.9% less job time and scored 0.0851 higher. Guided was both faster in cumulative job time and higher-scoring than Brief-errors on seven of the 15 paired tasks.

The job counts add information about the character of this expenditure.

Across the common tasks, Guided submitted 789 recorded quantum-chemical jobs, Standard 419, Record 431, and Brief-errors 883. Recorded failed-job counts were 114, 45, 62, and 167, respectively, with cancellations recorded separately. The four difficult studies NU3, NU5, NU6, and NR1 accounted for 163.5 job-hours across their 16 runs, 24.3% of the 673.82 job-hours in the available 63-run effort archive. These data locate a substantial part of computational expenditure in studies where additional attempts did not uniformly complete the missing network. In the archived Standard missing-graph cases, calculations could also outlive the research session, leaving subsequent results outside an active study. Continuing the research after a calculation returns is therefore part of the delivered workflow measured here.

Guided exceeded Brief-errors by 0.0388 overall, with an interval of $[-0.0317, 0.1076]$, 26 wins, ten losses, and 14 ties. The comparison varied by research demand: Guided's mean advantage was 0.1514 on Normal and 0.1295 on Hard, whereas Brief-errors had the higher Impossible mean. Together with the common-task effort comparison, these observations support examining what a diagnostic enables the system to repair: a specific saddle assignment, a path endpoint, an energy reference, or the judgment that the permitted method cannot support the requested claim. More diagnostic text does not impose the same ordering across these different decisions.

3.1.6 Per-task patterns identify the chemical relations that need to persist

Figure 10 places all 50 tasks and four configurations alongside their complete score components. The resulting landscape shows both consistent gains and specific reversals. Guided performed strongly across the Hard comparative studies, while NB1 and NI1 show that multistep studies could also retain high scores and strong evidence support. NB1 scored 0.9628 under Guided, 0.9128 under Standard, and 0.9900 under Brief-errors; Guided and Brief-errors both had self-consistency 1. NI1 scored 0.9958 under Guided and 0.6196 under Standard, with self-consistency 1 and 0.35. In contrast, NQ5 favored Record over Guided, and NU3 favored Brief-errors. These differences make the individual target steps and evidence relationships more informative

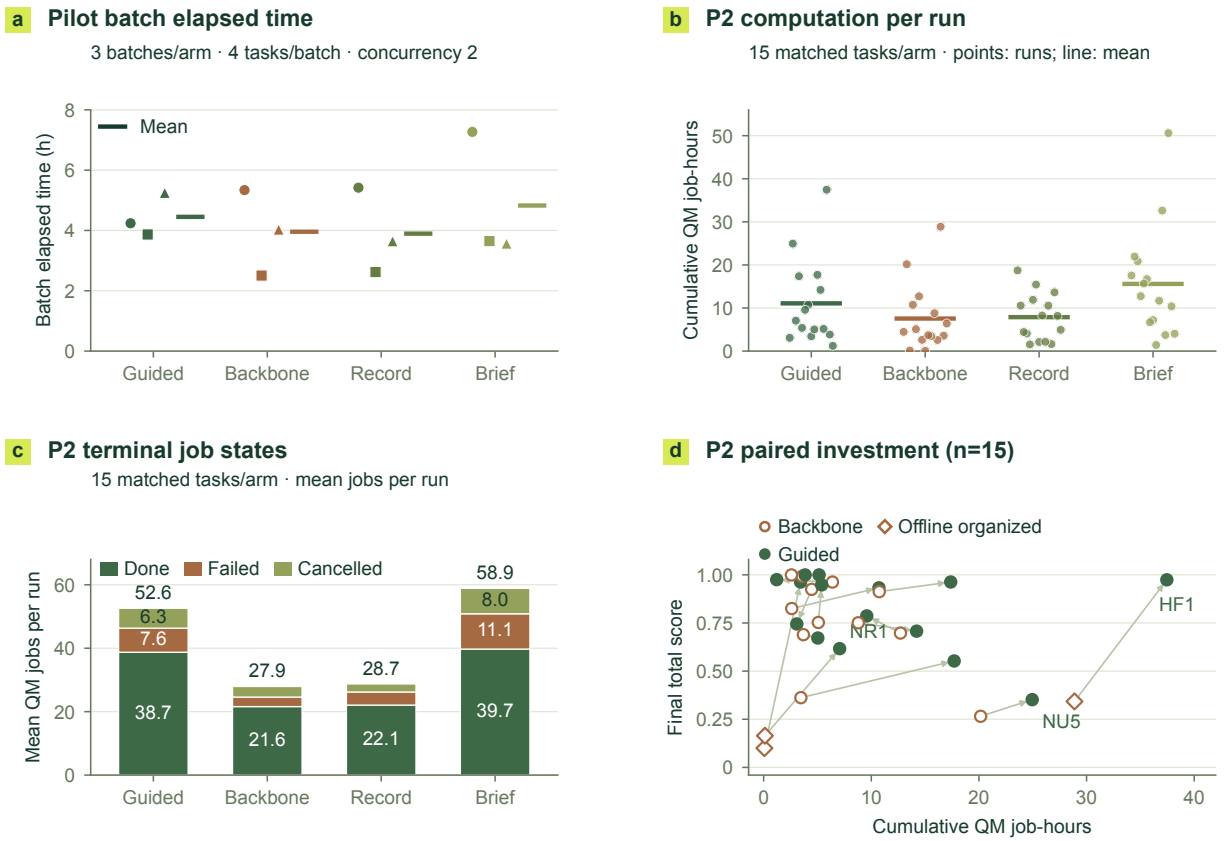


Figure 9. Research elapsed time and computational investment. (a) Twelve P1b four-task batches at concurrency two, three per arm; lines mark means. (b) Cumulative QM process durations for the 15 P2 tasks shared by all arms. (c) Mean completed, failed, and cancelled jobs in that cohort. (d) Same-task Backbone-to-Guided changes in investment and final score; diamonds mark offline-organized Backbone graphs. Job-hours sum process walltime; panel a measures batch latency. Backbone/M1 effort is reconstructed from native CPU time divided by four.

than a single ordering by network length.

Three archived comparisons clarify which chemical relations matter (Figure 11). In HA8, Guided reported 29.46 kcal/mol and Standard 20.43 kcal/mol against a reference of 29.48 kcal/mol. The archive traces this 9.03 kcal/mol difference to the energy reference: the lowest all-s-trans reactant minimum in Guided and a reactive cisoid conformer in Standard. Local IRC connectivity and the stable-species reference for an activation energy consequently need to remain associated without being conflated.

The former identifies the structures reached by a particular path; the latter determines the energetic starting point specified by the research question.

NU6 exposes a related issue in path identity. Guided's recorded concerted saddle had an energy of -247.030042392 Ha and the betaine minimum -247.060060699 Ha, giving a difference of 18.836772 kcal/mol. Guided described this as an energy cost involving the concerted saddle region while retaining the target betaine-to-product ring-closure saddle as unresolved. Standard delivered 18.837 kcal/mol as the second-step ring-closure barrier. The numerical fingerprint thus locates the divergence in what scientific step an energy difference was taken to represent. Guided's attempts from interpolated betaine structures and constrained scan peaks repeatedly relaxed toward the direct reactant-to-product saddle region. The requested path identity had to be checked again after optimization, rather than being inherited from the name of its initial guess.

These cases complement the HH1 evidence comparison: the same research state must preserve which conformer defines the reference, which stationary point was actually found, which endpoints it connects, and which claim those results support. The Easy-tier comparisons further distinguish the chemical identities declared in the final mechanism graph from the physical calculations and numerical quantities retained in the run.

In G2, Guided scored 0.2389 and Standard 0.7769. Guided's native logs supported an approximately 0.845-kcal/mol double-H-transfer barrier. After three rejections of numerical barrier declarations on a companion edge, the agent removed the numerical fields. Supervisor feedback prompted restoration, but a fourth rejection and a subsequent saddle-evidence check preceded their final removal, together with the saddle energy/log bindings. The final graph retained the values in text and received accuracy 0. Brief-errors retained the corresponding numerical result and scored 0.8500.

The accuracy values of 0.7 in D1, D3, and L2 followed the specified cap on numerical-deviation penalties. D1's barrier difference from Standard traced to the reactant-conformer reference; D3's mainly reflected different saddle energies. L2 distinguished reactive cZc and lower tZt references. Standard delivered both associated barriers, 18.99 and 28.28 kcal/mol, but reference matching selected the former. The Supplementary Information provides the final workflow-

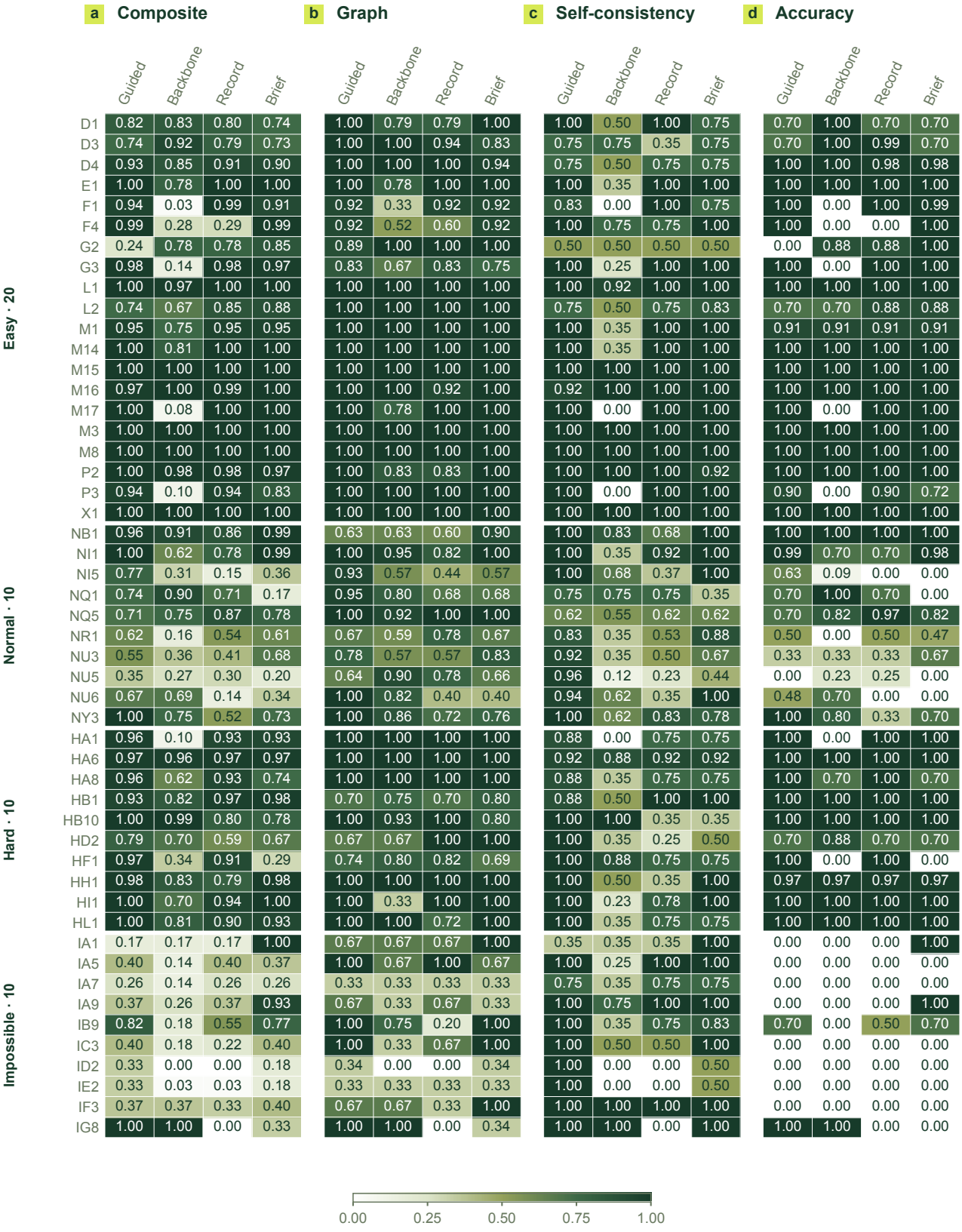


Figure 10. Complete CloudraBench results for all 50 tasks and four configurations. The aligned blocks show composite, Graph, Self-consistency, and Accuracy on the same 0–1 scale; all 800 cells are present. Values are rounded to two decimals; the source table retains full precision.

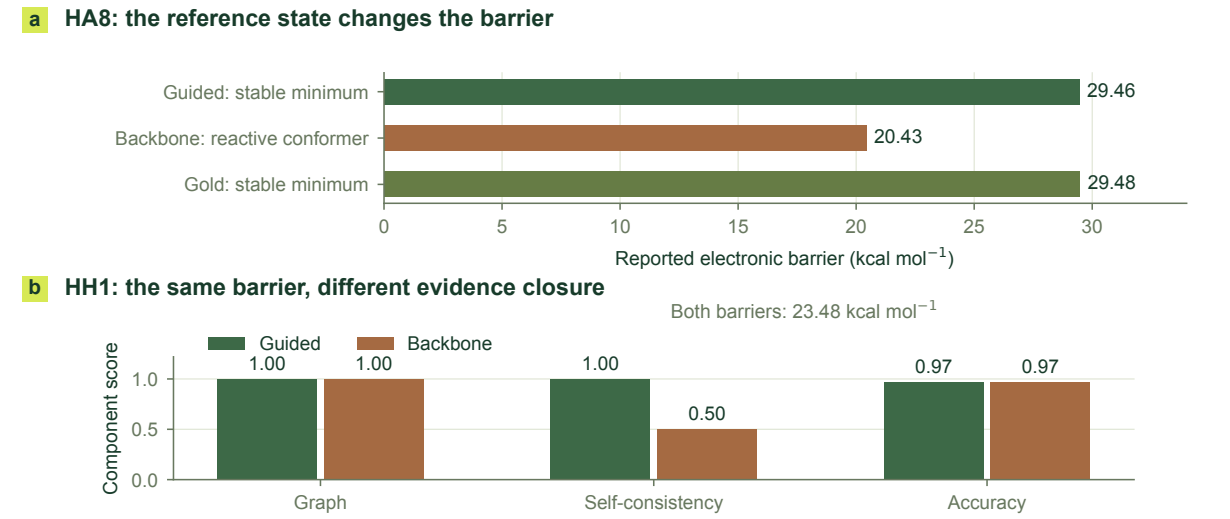


Figure 11. Scientific attribution in matched Guided–Backbone tasks. (a) HA8 barriers use stable-minimum or reactive-conformer references. (b) HH1 retains the same barrier and accuracy, with different evidence closure. (c) NU6's recorded energy difference is assigned to different pathways. Values and assignments derive from the archived records.

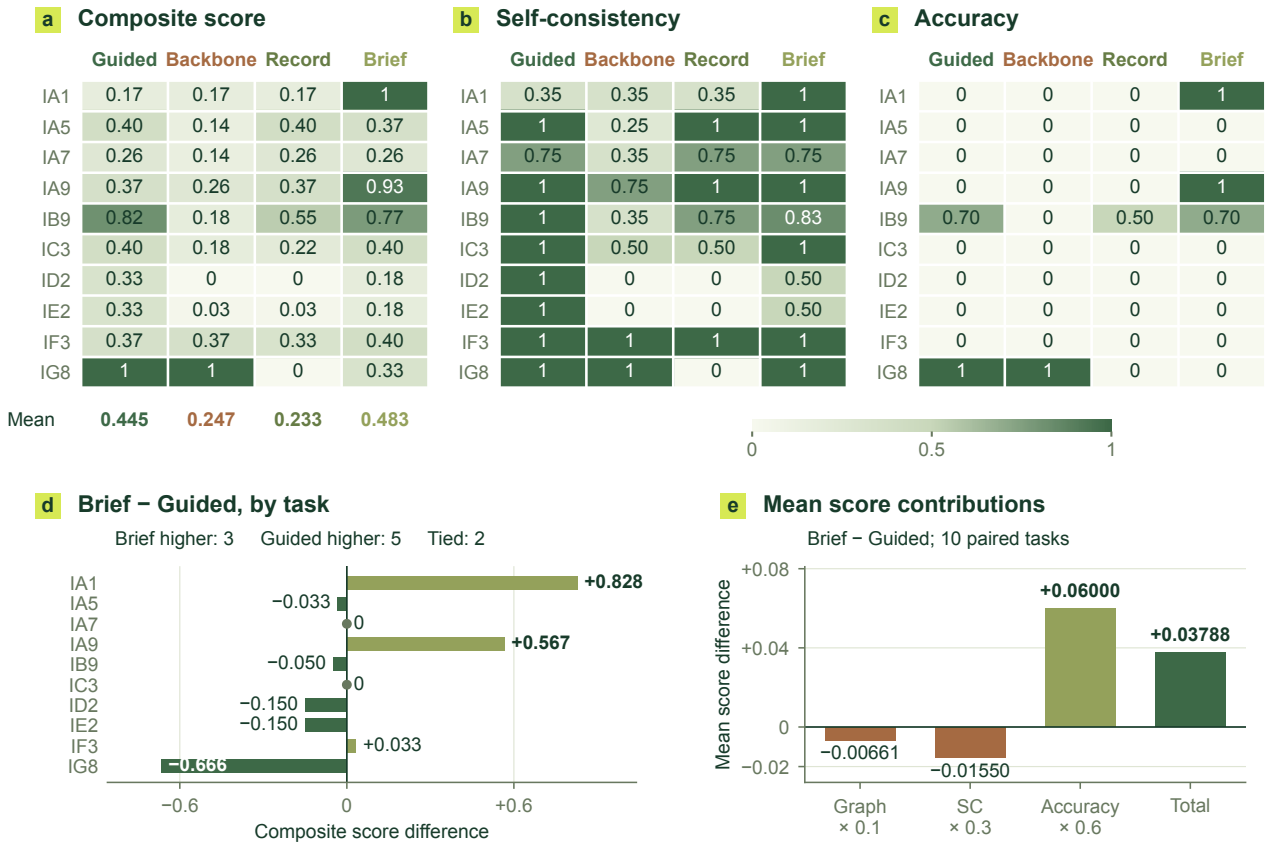


Figure 12. Impossible-tier outcomes and the brief-diagnostic ablation. (a–c) Composite, self-consistency (SC), and accuracy for all ten tasks and four arms, using the complete organized scores on a common 0–1 scale. Backbone denotes the actual same-backbone Standard arm. (d) Paired Brief-minus-Guided differences: the mean advantage of 0.03788 is concentrated in IA1 and IA9, with an opposite outcome in IG8; Guided scores higher on five tasks, Brief on three, and two tie. (e) The mean difference decomposes into weighted accuracy (+0.06000), Graph (−0.00661), and SC (−0.01550); the small discrepancy in their sum reflects rounding of saved totals. The paired task-bootstrap 95% interval is −0.188 to +0.284; task-specific claims are analyzed below.

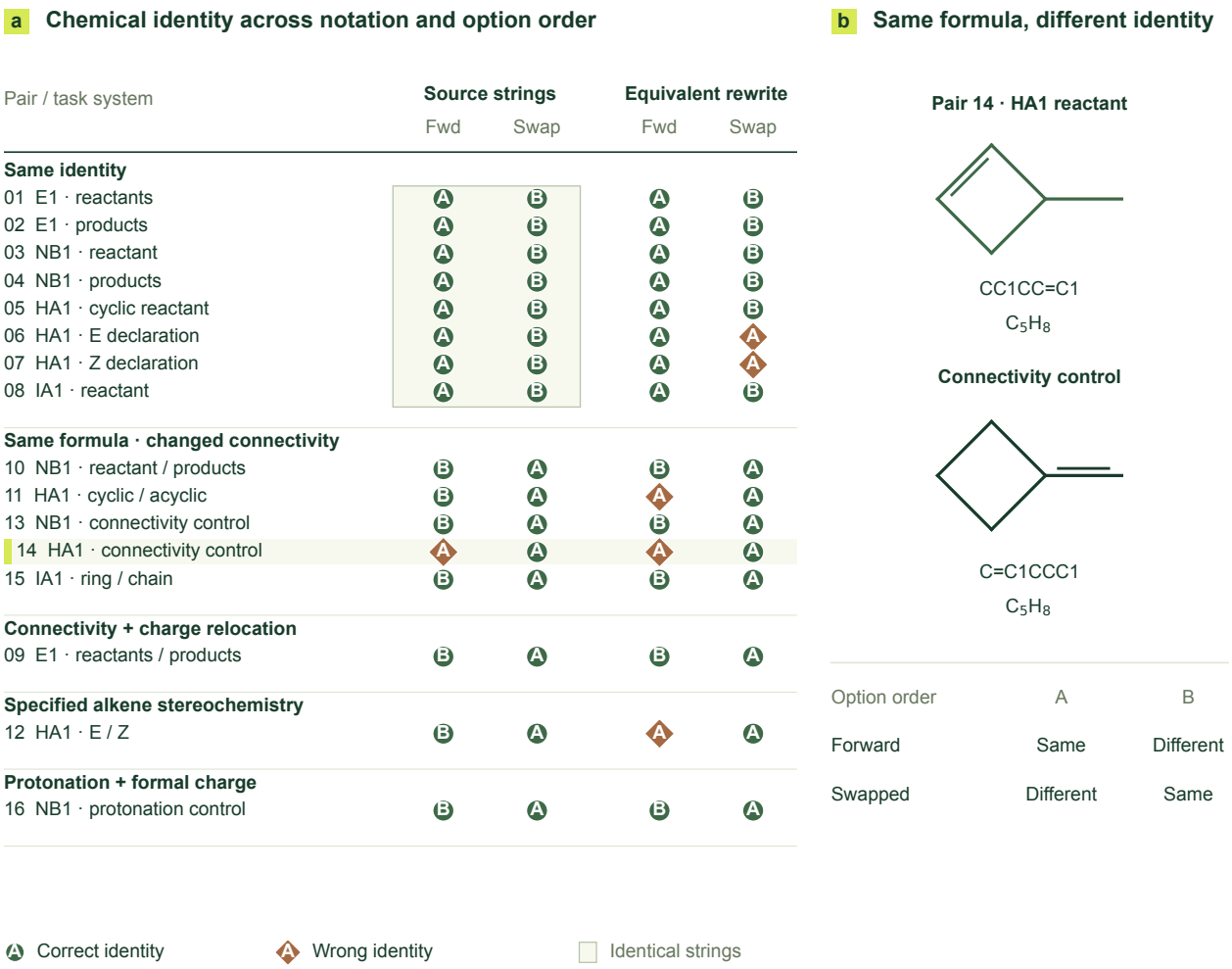


Figure 13. Molecular-identity judgments under changes in molecular representation and option order. Each of the 16 molecular pairs has four input conditions. Circles and diamonds indicate correct and incorrect judgments, respectively; the letters retain the model's actual answers. Shaded regions denote identical-string conditions. The right panel shows the pair-14 connectivities `CC1CC=C1` and `C=C1CCC1`, both C_5H_8, and the meanings of the answer options.

level components, native evidence, and delivery records for these five tasks.

3.1.7 Impossible tasks distinguish local support from the requested negative conclusion

The Impossible tier tests whether research under xTB/PM7 reaches the specified method-limited conclusion. Brief-errors averaged 0.4829, Guided 0.4451, Standard 0.2473, and Record 0.2325 (Figure 12). Brief-errors exceeded Guided on three tasks and scored lower on five, with two ties. Its mean advantage of 0.0379 was concentrated in particular answers: IA1 contributed +0.8283 and IA9 +0.5666, whereas Guided's IG8 result contributed a difference of -0.6661 in the Brief-errors-minus-Guided direction.

In IA1, concerning cyclobutadiene's square-geometry automerization target, Brief-errors had graph, self-consistency, and accuracy scores of 1, while Guided's accuracy was 0. In IA9, concerning dioxetane thermolysis, Brief-errors and Guided both had self-consistency 1, but accuracy was 1 and 0, respectively. IG8 showed the opposite accuracy ordering for the chromium-dimer target. Across the seven tasks with the same accuracy score in Brief-errors and Guided, Brief-errors averaged 0.0500 lower. Its overall Impossible advantage decomposed into weighted contributions of -0.0066 from graph, -0.0155 from self-consistency, and $+0.0600$ from accuracy. The higher mean arose from the requested conclusions on specific tasks, while Guided retained stronger local support on several others.

Across all 40 Impossible runs, four had accuracy 1 and 14 combined self-consistency 1 with accuracy 0. This separation is visible directly in the component matrices. For example, both Guided and Brief-errors obtained self-consistency 1 in IA5, IC3, and IF3 without receiving accuracy credit for the requested negative target. Guided and Brief-errors both received partial accuracy of 0.7 in IB9, whose requested network contains positive and negative components. These outcomes retain the distinction between local computational support and the target claim.

The saved Brief-errors IC3 graph provides a concrete example of this distinction in scientific reporting. It marks the lowest open-shell singlet and its state-change edge inaccessible and withholds the requested singlet–triplet gap. The computed closed-shell singlet remains as a separate characterized branch, and the two method-dependent relative energies are explicitly excluded from the requested gap. The run therefore retained useful exploratory results while declining to promote them into an answer about a different electronic state. Its final accuracy score remained 0. The claim and its

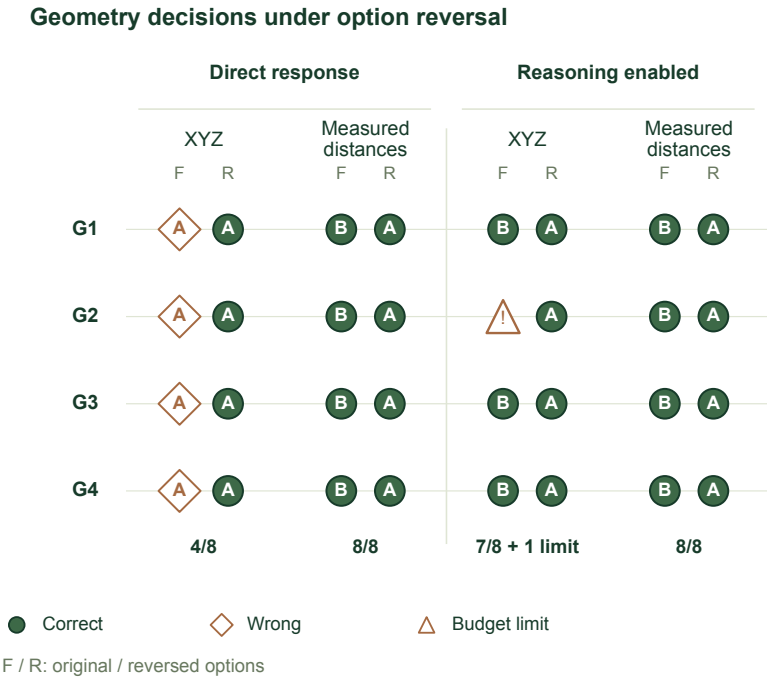


Figure 14. Geometry judgments under coordinate inputs and measured-distance inputs. Four source structures are compared under direct response and reasoning-enabled conditions, each with original (F) and reversed (R) option order. Circles indicate correct judgments, diamonds incorrect judgments, and the triangle a response that reached the output budget. Letters retain the actual answers.

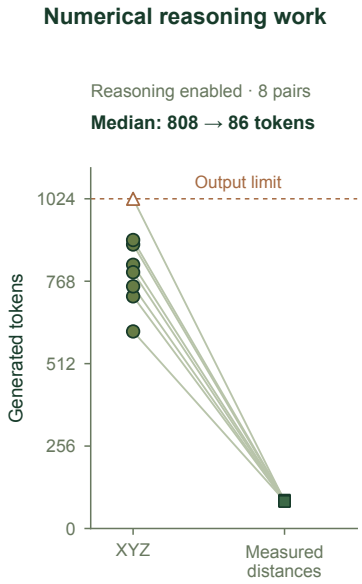


Figure 15. Generated-token counts with reasoning enabled. Each line connects Cartesian-coordinate input and measured-distance input for the same source structure and option order; eight paired comparisons are shown. The dashed line marks the output limit.

final score are both reported here, with target-object assignment considered in the Discussion. This behavior complements the IA1, IA9, and IG8 score reversals by showing precisely which result was accepted as computable and which target claim remained unsupported.

3.2 From chemical intent to reliable geometric operations

The end-to-end benchmark evaluates research capability across complete tasks. The following sections examine the semantic correspondences involved in chemical identity, target-specific geometric operations, evidence associations, and completion decisions. Archived research trajectories locate concrete problems; controlled local studies with Qwen3.8:27b use paired inputs and internal-state interventions to examine how particular information changes local judgments and actions.

3.2.1 Molecular identity provides a persistent reference for operations

A geometric modification first requires identifying the molecule, atoms, and reaction center to which it applies. A molecule can be represented by different SMILES traversals, while a molecular formula can describe different connectivities. To examine how the model handles these distinctions, we constructed 16 molecular-identity pairs covering identical identities, connectivity changes within the same molecular formula, changes in connectivity and charge location, double-bond stereochemistry, and protonation changes. Each pair was presented using the original strings or equivalent rewritings and with two answer orders, yielding 64 judgments.

The model answered 58 correctly. Equivalent rewriting changed the semantic judgment in 4 of 32 matched representation comparisons; reversing answer order changed the judgment in 6 of 32 matched option-order comparisons. All six of the latter cases involved repeated selection of the letter A. One C_5H_8 pair in Figure 13 illustrates this behavior directly: an endocyclic and an exocyclic double bond define different connectivities, yet the model answered A under all four conditions, although A meant "same" in one order and "different" in the other. This result locates representation sensitivity in judgments about the chemical objects themselves and indicates the need for stable identity references throughout a research process.

Cloudra assigns this role to molecular topology, atom mapping, and reaction centers. These references allow subsequent measurements, fragment movements, and endpoint comparisons to remain attached to the originally specified objects as the model moves between descriptions and configurations. Atom correspondence also connects product-endpoint alignment, path interpolation, and reaction-center monitoring, allowing local geometric operations to proceed with a consistent set of chemical identities.

3.2.2 Geometric measurements reduce repeated numerical reasoning

After identifying the research objects, the model must extract spatial relationships relevant to an operation from their coordinates. We selected four actual structures from the NB1 trajectories and asked the model to compare two specified interatomic distances. Each structure was presented with both answer orders and with either a complete XYZ geometry or program-measured distances. These two information conditions were further crossed with direct-answer and thinking modes, producing 32 native chat calls (Figure 14).

In direct-answer mode, all eight responses to XYZ inputs selected A, with four correct answers. Providing measured

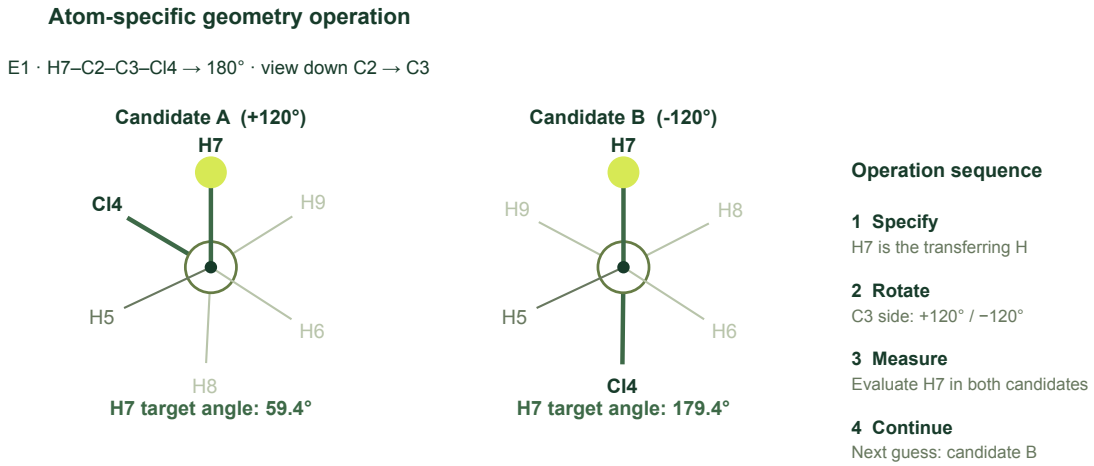


Figure 16. Target-specific geometry operations in E1. Newman projections retain the directions of the original candidate coordinates viewed down C2→C3. H7 is the specified transferring atom. Candidate A and candidate B give target angles of 59.4° and 179.4°, respectively. The operation sequence is shown alongside the projections.

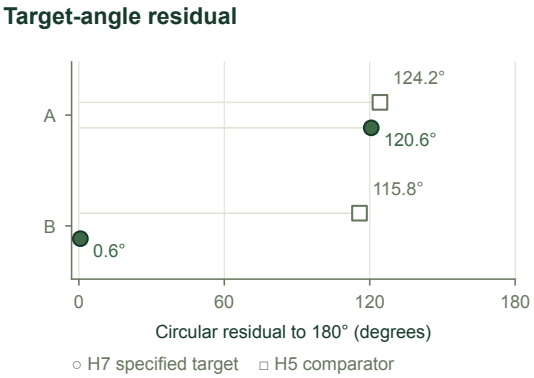


Figure 17. Circular angular residuals from the specified 180° target for candidates A and B. H7 is the specified target atom; H5 is the comparator.



Figure 18. Next-action choices under three local option permutations. When only H5 measurements are supplied, the model requests H7 measurements; with H7 supplied, it selects candidate B.

distances produced eight correct answers and preserved the semantic answer when option order was reversed. With thinking enabled, the XYZ condition produced seven correct answers; the remaining response reached the output limit of 1,024 generated tokens. Increasing the budget for that input alone yielded a correct answer within 1,353 generated tokens.

All eight answers in the measured-distance condition were correct. Across the eight paired comparisons with thinking enabled, the median generated token counts were 807.5 for XYZ inputs and 86 for measured distances.

These results suggest that geometric judgment combines atom identification, numerical calculation, and relational comparison. Additional reasoning can perform these local calculations, while program measurements directly supply the relational quantities needed for the subsequent decision. Cloudra therefore allows the model to express objectives through bond lengths, bond angles, dihedral angles, and fragment relationships, and uses deterministic operators to change the coordinates. This division of work assigns repeated numerical operations to the program and makes the resulting changes directly inspectable, consistent with the allocation of responsibilities in program-aided language models.[38] Descriptions of steric crowding, reactant preorganization, and local fragment rotation build on this foundation to use spatial relationships in selecting and constructing candidate configurations.

3.2.3 Feedback must match the specified atoms and geometric objective

Deterministic execution must remain aligned with chemical intent. An actual E1 trajectory provides a complete example. The model sought to bring the dihedral angle defined by the transferring hydrogen H7, C2, C3, and leaving atom Cl4 close to 180°. It rotated the fragment on the C3 side about the C2–C3 axis by +120° and −120°, producing candidates A and B, and set the C3–Cl4 distance to 2.35 Å. The representative dihedral in the default structure report involved H5, whereas the current objective involved H7. The model then explicitly measured H7–C2–C3–Cl4 in both candidates, obtaining approximately 59.4° and 179.4°, and used candidate B in its next actual submission.

Figure 16 shows Newman projections along C2→C3 drawn directly from the saved candidate coordinates, and Figure 17 summarizes the corresponding target-angle residuals. Candidate B places the H7 target angle within 0.6° of 180°, compared with a residual of 120.6° for A.

The corresponding residuals for the H5 comparison were 124.2° and 115.8° in A and B, respectively. These measurements reveal the object dependence of feedback: a reading

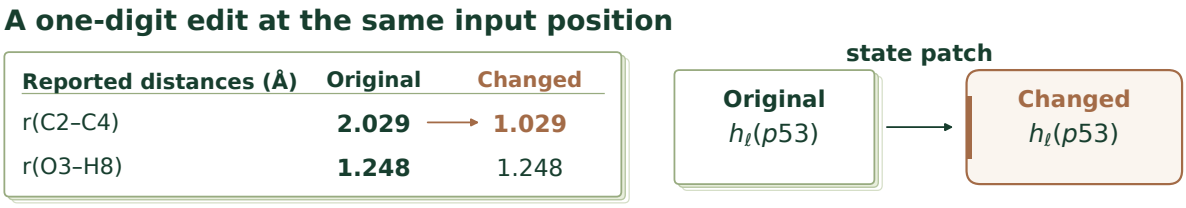


Figure 19. Intervention on a single reported distance. A one-digit edit changes the C2–C4 distance from 2.029 to 1.029 Å while O3–H8 remains 1.248 Å. The original-input residual state at token position 53 is inserted into the changed-input run at a specified layer. Token positions are zero-based.

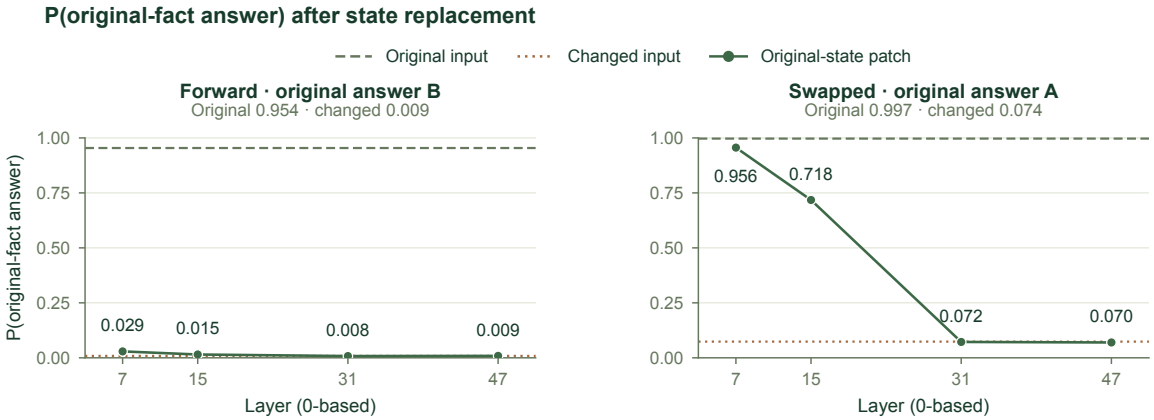


Figure 20. Original-fact answer probability after state replacement. Curves show full-vocabulary probability mass for the original-fact answer under forward (B) and swapped (A) option order, including bare and leading-space token spellings. Horizontal lines show original-input and changed-input baselines. Layer indices are zero-based.

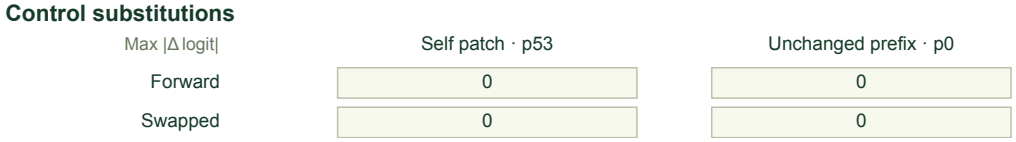


Figure 21. Control substitutions for the state-intervention experiment. Same-state self replacement at p53 and replacement of the unchanged prefix at p0 preserve all vocabulary logits in both option orders. The table reports the maximum absolute logit difference. Token positions are zero-based.

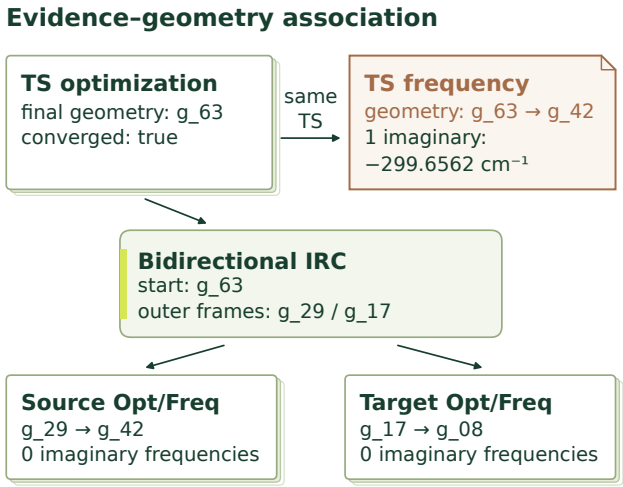


Figure 22. Evidence–geometry association in E1. The frequency geometry reference changes from g_63 to g_42 while local termination, convergence, energy, and frequency values remain fixed. Arrows retain the TS–IRC–endpoint optimization/frequency chain.

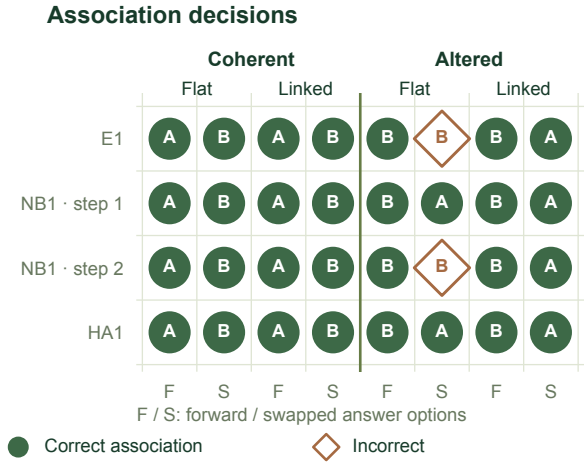


Figure 23. The 32 association judgments cross four source chains, coherent/altered links, flat/linked presentation, and forward/swapped answer order. Saved A/B responses and correctness markers are retained.

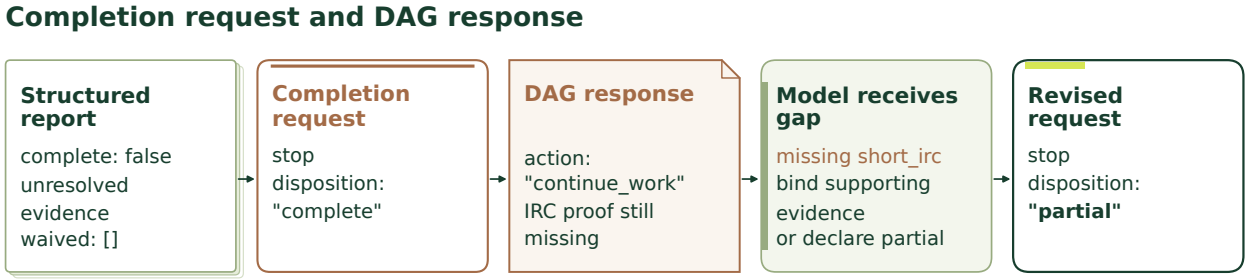


Figure 24. Completion requests and DAG feedback in the original IA1 trajectory. A report with unresolved evidence is followed by a completion request, a response identifying missing short_irc proof, feedback to the model, and a revised partial-completion request.

acquires its chemical meaning through its correspondence to the currently specified atoms and operational objective.

Local comparisons using the same measurements further connect information to action. When only H5 measurements were provided, the model requested an additional H7 measurement under all three cyclic option permutations. Adding H7 measurements led it to select B under all three permutations. Two further paired tests of information sufficiency also shifted from requesting additional measurements to judging the information sufficient when the target quantity was added. Atom mapping, internal coordinates, fragment operations, and feedback can therefore be understood as one sequence of research actions: identify the target object, make the geometric change, measure the target quantity, and select the next candidate. Preorganization, rigid-body alignment, and path preparation use the same correspondence to carry reaction intent into subsequent exploration of the potential energy surface.

3.2.4 Internal transfer of measured facts affects selection

A controlled local comparison isolates how a supplied geometric fact enters an immediate choice. It complements the trajectory analysis of how measurements are retained through geometric operations and later decisions. To examine how the model uses supplied measurements, we performed activation replacement at a single token position in the same local quantized model.[39] One distance in the input was changed from 2.029 Å to 1.029 Å, while the other remained at 1.248 Å, reversing the correct comparison answer. We then transferred the residual-stream state at the numeric token position from the original input into the modified input at layers 7, 15, 31, and 47, and read the full-vocabulary probability assigned to the answer supported by the original fact.

The effect varied with layer and option order (Figure 20). With reversed options, the probability of A, the original-fact answer, increased from 0.074 for the modified input to 0.956 and 0.718 after replacement at layers 7 and 15, respectively. Replacements at layers 31 and 47 yielded 0.072 and 0.070. With the original option order, the probability of answer B changed from 0.009 to 0.029 and 0.015 after replacement at the first two tested layers, while the last two remained close to the modified-input baseline.

Self-replacement controls and control replacements at an unchanged prefix position produced zero logit differences across the full vocabulary.

Changing the residual state at one numeric position changes the subsequent choice in this probe. The difference between option orders is consistent with the positional sensitivity observed in Section 3.2.1. Together with the measurement and association controls, these results motivate keeping geometric quantities linked to their chemical objects and operations so that later choices can be checked against the same recorded facts.

3.3 From local computational results to evidence-supported scientific delivery

3.3.1 The DAG preserves scientific support relationships among results

Once geometric objects enter computation, optimized structures, energies, frequencies, and paths must be assembled into support for the same reaction step. The local validity and attribution of each result jointly determine this support chain: frequencies must correspond to the declared saddle point, the IRC must start from that saddle point, and optimizations and frequencies for the final frames on both sides must correspond to the respective endpoints. The DAG records these relationships on species and elementary steps, providing a persistent research index as results accumulate. Execution state, numerical convergence, and scientific support are recorded separately, allowing subsequent work to wait for a calculation, revise and rerun it, or continue assembling evidence.

We extracted four source evidence chains from the archive and constructed versions that preserved the links or changed a single reference. Crossing these with flat and linked representations and two answer orders yielded 32 inputs. The altered references concerned the frequency geometry, IRC starting geometry, endpoint-optimization input, or target species; local termination, convergence, energy, and frequency values were held constant. Figure 22 illustrates the E1 case: changing the frequency geometry reference from the saddle point to an existing endpoint changes how that frequency result supports the current step.

All 16 judgments with linked representations were correct, compared with 14 of 16 with flat representations. Both errors occurred when a reference had been changed and the options reversed, in E1 and the second NB1 step. In subsequent native follow-up calls on two selected source chains, enabling thinking yielded correct judgments for all 16 judgments of the evidence relationships, including the two previously incorrect cases.

This result suggests that explicit links reduce the model's work in locating and assembling computational relationships, consistent with observations that graph encoding affects relational reasoning in language models.[40] In

Cloudra, this representation is further bound to actual computational artifacts, allowing scientific facts to accumulate continuously for the same species, step, and attempt.

3.3.2 Diagnostics turn evidence gaps into concrete research actions

Diagnostic feedback connects an evidence defect to the chemical object and calculation that require revision. Cloudra's structured diagnostics identify specific problems in quantum-chemical outputs, including steric-clash atom pairs, imaginary-mode misalignment, and disconnected trajectory endpoints. These details give the model a target for its next local calculation.

Two archived trajectories illustrate this connection. In a Brief-errors HA1 trial, compact error codes replaced the detailed IRC-endpoint linkage diagnostics. The Computational Agent repeatedly substituted source-minimum references while the geometric disconnection remained unresolved. In a Guided NB1 trial, the verification found zero imaginary frequencies and returned a normal-mode diagnostic. The Coordinator requested a retry with revised transition-state guess parameters and optimization constraints, which the Computational Agent used in the rescheduled calculation. These examples show how a diagnostic can connect a failed check to the objects and parameters that need attention.[41]

The roles operate on the same recorded chemical entities. Computational Agents refine the geometries and paths for their assigned steps; the Coordinator manages hypotheses, branches, and route selection; and the Supervisor reviews the accumulated support against the task rubrics. Central asynchronous execution and durable ledgers preserve the calculation records and shared context used by these roles throughout exploration.

3.3.3 Outstanding requirements must remain part of completion decisions

An archived IA1 trajectory shows how a completion request can conflict with the recorded scientific requirements. The reaction step still lacked mandatory bidirectional short_irc verification, and the waiver list was empty. The Supervisor had earlier commented positively on intermediate optimization convergence. The model subsequently requested completion while the IRC requirement remained unresolved.

The harness rejected that request and returned the outstanding short_irc requirement to the model. In the next submission, the model changed its status to partial completion (Figure 24). The recorded requirement remained in force through both the positive comments on local calculations and the later completion request.

We also tested whether the model could identify the outstanding IA1 requirements from the recorded state. All 12 judgments were correct (100%) across six pairs with reversed answer-option order, using both a minimal summary and an extended 12-message history. The model identified the unsatisfied requirements in each state query and option order. The Delivery DAG retains the verified entities and unresolved requirements that inform the choice between further calculations and a partial scientific deliverable.

3.4 Workflow guidance, scientific judgment, and outlook

The primary comparison shows that Cloudra can improve the same backbone's research delivery by keeping chemical objects, computational results, and scientific requirements connected throughout exploration. The consistent Hard-task improvement and the Normal-task gains place this contribution in comparative stereochemical studies and the completion of multistep research. The complete 50-task decomposition locates gains in both evidence closure and target accuracy; in Hard tasks, their weighted contributions are +0.1353 and +0.1274. HH1 and HI1 additionally show why evidence closure matters even after a reference-consistent barrier has been obtained: the numerical result and the path supporting its scientific attribution are separate achievements.

The archived cases suggest three requirements for an autonomous research process. First, the identity of a scientific object must persist beyond the current language description. A reactant species can contain several conformers, and a requested reaction step can produce an optimized saddle belonging to a different route. HA8 requires the energy denominator to remain associated with the stable species while the IRC retains its actual local endpoints. NU6 requires a newly optimized stationary point to be assigned from its resulting connectivity rather than from the intended name of the initial guess. These are global relations among results accumulated at different times, consistent with the role of external planning and verification feedback in language-model workflows.[42] A persistent research graph gives the model a place to inspect and update them as each calculation changes what is known.

Second, feedback must identify the level at which the research should change. A missing endpoint optimization calls for a different action from a saddle that repeatedly relaxes to another route, and neither is equivalent to a method that cannot describe the target electronic state. Language models generate plausible next actions from their current context; a chemically familiar description can remain coherent even when the numerical or path evidence has changed, paralleling documented gaps between generated explanations and the factors driving model answers.[43] External measurements and structured evidence can expose that mismatch, but the agent must still decide whether to repair a calculation, revise the pathway hypothesis, change a reference assignment, or withhold the requested conclusion. The Record comparison shows that making tools available is insufficient to ensure that these relationships remain active in the final argument. The Brief-errors comparison further shows that the useful content of feedback depends on the decision being made, complementing evidence on the importance of external feedback for reasoning correction.[41, 44]

Third, completing a local proof and recognizing the scope of a scientific claim need to be supported together. Brief-errors' higher Impossible mean came from particular target judgments in IA1 and IA9, with the reverse result in IG8. Its IC3 graph provides a more detailed behavioral example: the run retained a characterized closed-shell branch while refusing to treat it as the requested lowest open-shell singlet or to report its relative energy as the target gap. This distinction is a valuable form of scientific judgment. It preserves what exploration established without allowing a convenient result to

replace a different research question. The crossed outcomes motivate a task-dependent hypothesis: detailed diagnostics can help complete a valid local proof, while continuing to repair that proof may be unhelpful when the method or target itself needs reconsideration. The present final-result comparisons do not establish the causal effect of diagnostic length or measure the internal cognitive mechanism responsible.

The Easy-tier G2 case exposes a workflow interaction: blocking numerical declarations before the companion edge passed its checks led the agent to remove quantities already supported by native calculations. Supervisor feedback restored those fields, but further rejection preceded their deletion from the final graph. The distinction needed here is between preserving a provisional observation and accepting the mechanistic claim attached to it. A validation status should qualify an archived quantity while retaining the observation and its provenance. This trajectory distinguishes the retention of a supported observation from acceptance of the associated mechanistic claim; its partial self-consistency contribution also remains distinct from an accuracy failure.

The identity of a scientific claim cannot be inferred solely from the molecular connectivity of its endpoints. Record/IA5 correctly rejected the requested disrotatory path, but the archived adjudication records that a matcher selected another positive path with the same endpoints. Brief-errors/IC3 received accuracy 0 despite explicitly distinguishing the requested open-shell state from its separate closed-shell branch. The final scores are interpreted together with the saved claims. Evaluation must preserve pathway and electronic-state distinctions when assigning results to the requested target, just as the research workflow must.

A conditional target-assignment analysis in the Supplementary Information applies the existing negative-credit rule to the intended IA5 and IC3 objects while holding their Graph and Self-consistency values and the other 198 runs fixed. Each assignment changes one composite from 0.4 to 1.0, increasing its workflow's Impossible mean by 0.0600. With both assignments, Brief, Guided, Record, and Standard have conditional means of 0.54294, 0.44506, 0.29250, and 0.24733. The ordering among Brief, Guided, and Record is unchanged; Record moves above Standard. Every Guided–Standard pair remains unchanged. The manuscript reports the final evaluation scores; the conditional table isolates object assignment from the agent's final scientific judgment.

P2 contains one retained run per task and configuration. Its task-cluster bootstrap intervals describe variation across the sampled task mix, not stochastic variation from independent runs of the same task. The separate P1b study supplies three repeats on each of four tasks under each configuration; these 48 runs do not provide repeated measurements for every task in the 50-task P2 study. The complete organized P2 table covers all 200 runs and their three score components. Some Hard-task accuracy targets concern qualitative channel identity, so the reported gain includes pathway identification and evidence completion as well as numerical barriers. Repeating the full task set would characterize run-to-run variability and the frequency with which each research route is selected.

The Impossible tier is defined by the allowed-method contract, rather than a universal claim that the corresponding chemistry cannot occur. Its component values also use negative-target rules: self-consistency 1 on an explicit negative conclusion is distinct from a positive TS–IRC–endpoint chain. The 14 Impossible cases with self-consistency 1 and accuracy 0 should consequently be interpreted through their target claims and provenance. NU6's matching energy fingerprint identifies a related problem for accessible pathways: an energy difference can acquire different scientific meanings when its saddle identity and the requested step are not maintained together. These cases motivate persistent checks of what each calculation actually answers as the research hypothesis changes.

Timing and effort are likewise reported at their observed units. The 12 pilot batches measure real elapsed time for four tasks at concurrency two, including the original failures and stopping outcomes. The P2 effort archive covers selected deep-dive studies, with 15 tasks shared across configurations; cumulative job-hours represent recorded quantum-chemical process durations, with a documented native-log reconstruction for Standard/M1. They are distinct from core-hours, financial cost, and end-to-end P2 research latency. Concurrent calculations shared computational resources, and CPU queueing may have delayed individual results. The measurements therefore support a comparison of the observed workflows and investments, while further separating queueing, calculation, and agent-active time would clarify where elapsed time is spent.

The delivery cases identify a concrete systems direction alongside the scientific one. Waiting for a calculation, receiving its completion event, resuming the research, and assembling the final artifact must share a continuous task lifecycle. Standard's missing graphs followed premature completion while calculations were still pending; Guided/HI1 also exposed a notification that did not wake a waiting session. Preserving an unfinished goal across these transitions allows newly completed computations to become usable research information. At the scientific level, future work should make repeated returns to the same stationary-point region visible and help the agent decide when another local attempt is unlikely to resolve the target pathway. Better candidate selection and explicit revision of research hypotheses can then complement the improved preservation of evidence.

Cloudra's default preorganization samples 32 rigid relative placements, applies local MMFF94 relaxation, and selects candidates by energy and clustering. This candidate-generation procedure samples local arrangements. A frequency-verified minimum is a stationary-point qualification; it does not establish that all lower conformers have been searched. HA8 shows why the specified stable reference and the local reactive conformer must both be retained, while a global-minimum claim requires a separate assessment of conformer coverage. CREST is available through a separate optional backend; the 200-run study does not establish uniform CREST coverage. Highly flexible systems may require broader sampling and more quantum-chemical calculations. Further work should measure conformer coverage and improve the allocation of calculations among competing candidates.

The benchmark compares electronic energy differences. It does not add zero-point, thermal/entropic, or standard-state corrections, and it does not quantify the free-energy contribution of solvation. This scope matters for NI5, specified in weakly acidic aqueous solution, and NU5, specified in dichloromethane at high concentration. Their scores as-

sess the prescribed electronic-energy references. For association and dissociation, entropy and concentration-dependent standard-state terms enter $\Delta G^{\ddagger}$; solvent participation can also change the chemical model required to describe a pathway. Inferring solution-phase rate control or stereoselectivity would require a consistent free-energy treatment of the relevant species and pathways. No solvent-sensitivity comparison was performed, so the present results do not determine the magnitude or direction of a solvent effect.

The atom-mapping coefficients and ambiguity window, and the decay, depth, and congestion settings in the branch-crowding descriptor, were fixed across the workflow comparisons. We did not vary them in a parameter sweep or measure each setting's separate contribution. The imaginary-mode alignment ratio of 0.5 is advisory; the stationary-point and IRC/endpoint evidence determine scientific acceptance. All P2 configurations used DeepSeek-v4.1-flash (1017), with the API alias `deepseek-v4-flash` and the recorded request settings. Comparisons with additional backbones would test how the workflow gains carry across models.

Our longer-term goal extends beyond small-molecule mechanism calculations to a broader range of computational chemistry. Support for multiple computational engines, shared execution infrastructure, and persistent scientific records already provides a basis for that expansion. Extending coverage will require appropriate representations, methods, and validation criteria for each new class of problem. We aim for Cloudra to become a powerful research tool for computational chemists and a computational specialist for experimental teams. Cloudra will be released as open-source software under the GNU Affero General Public License (AGPL), supporting inspection, reuse, and community development of the system.

Data availability

The retained P2 benchmark records, the 48-run P1b study, Gold reference-calculation evidence, and supplementary reproduction materials are publicly available at `https://drive.google.com/drive/folders/1yZo3eJuCUuAgedQztgCtLyJcc8MZipf7`. Package contents, versions, and checksums are described in Supplementary Information Section S7 and the accompanying download records.

References

[1] K. Fukui. Formulation of the reaction coordinate. *The Journal of Physical Chemistry*, 74:4161–4163, 1970. doi: 10.1021/j100717a029.

[2] S. Maeda, Y. Harabuchi, Y. Ono, T. Taketsugu, and K. Morokuma. Intrinsic reaction coordinate: Calculation, bifurcation, and automated search. *International Journal of Quantum Chemistry*, 115:258–269, 2015. doi: 10.1002/qua.24757.

[3] Yunheng Zou et al. El Agente: An autonomous agent for quantum chemistry. *Matter*, 8:102263, 2025. doi: 10.1016/j.matt.2025.102263.

[4] Jinming Hu, Hassan Nawaz, Yi-Fan Hou, Lijie Chi, Xin-Yu Tong, Yuting Rui, Yuxinxin Chen, Arif Ullah, and Pavlo O. Dral. Aitomia: An Agentic Framework for AI-Driven Atomistic and Quantum Chemical Simulations. *Journal of Chemical Theory and Computation*, 22(13):6557–6571, 2026. doi: 10.1021/acs.jctc.6c00591. URL `https://doi.org/10.1021/acs.jctc.6c00591`.

[5] Jiaru Bai, Abdulrahman Aldossary, Thomas Swanick, Marcel Müller, Yeonghun Kang, Changhyeok Choi, Naruki Yoshikawa, Zijian Zhang, Jin Won Lee, Tsz Wai Ko, Aiwei Yin, Mohammad Ghazi Vakili, Chris Crebolder, Varinia Bernales, and Alán Aspuru-Guzik. El Agente Gráfico: A Semantic Execution Runtime for Scientific Agents, 2026. URL `https://arxiv.org/abs/2602.17902v2`. Version 2, 7 August 2026.

[6] Changhyeok Choi, Yunheng Zou, Marcel Müller, Han Hao, Yeonghun Kang, Juan B. Pérez-Sánchez, Ignacio Gustin, Hanyong Xu, Andrew Wang, Mohammad Ghazi Vakili, Chris Crebolder, Alán Aspuru-Guzik, and Varinia Bernales. El Agente Estructural: An Artificially Intelligent Molecular Editor, 2026. URL `https://arxiv.org/abs/2602.04849`.

[7] Dong Li, Sixuan Mi, Zihao Ye, Huan Xiong, Tao Xu, Tong Zhu, Aijia Zhang, Junqi Gao, Kaiyan Zhang, Shijie Wang, Bowen Zhou, Yuqiang Li, and Biqing Qi. Autonomous Chemical Mechanistic Discovery through Agentic Reasoning and Validation, 2026. URL `https://arxiv.org/abs/2609.11147`.

[8] Juan B. Pérez-Sánchez, Yunheng Zou, Jorge A. Campos-Gonzalez-Angulo, Marcel Müller, Ignacio Gustin, Andrew Wang, Han Hao, Tsz Wai Ko, Changhyeok Choi, Eric S. Isbrandt, Mohammad Ghazi Vakili, Hanyong Xu, Chris Crebolder, Varinia Bernales, and Alán Aspuru-Guzik. El Agente Quntur: A research collaborator agent for quantum chemistry. *Chem*, page 103259, 2026. doi: 10.1016/j.chempr.2026.103259. URL `https://doi.org/10.1016/j.chempr.2026.103259`.

[9] S. Riniker and G. A. Landrum. Better Informed Distance Geometry: Using What We Know To Improve Conformation Generation. *Journal of Chemical Information and Modeling*, 55:2562–2574, 2015. doi: 10.1021/acs.jcim.5b00654.

[10] Shuzhe Wang, Jagna Witek, Gregory A. Landrum, and Sereina Riniker. Improving conformer generation for small rings and macrocycles based on distance geometry and experimental torsional-angle preferences. *Journal of Chemical Information and Modeling*, 60(4):2044–2058, 2020. doi: 10.1021/acs.jcim.0c00025.

[11] RDKit. Getting Started in Python: Conformer Generation and Maximum Common Substructure, 2026. URL `https://www.rdkit.org/docs/GettingStartedInPython.html`. Online documentation; accessed 31 October 2026.

[12] Thomas A. Halgren. Merck molecular force field. I. basis, form, scope, parameterization, and performance of MMFF94. *Journal of Computational Chemistry*, 17(5–6):490–519, 1996. doi: 10.1002/(SICI)1096-987X(199604)17:5/6<490::AID-JCC1>3.0.CO;2-P.

[13] W. Kabsch. A discussion of the solution for the best rotation to relate two sets of vectors. *Acta Crystallographica Section A*, 34:827–828, 1978. doi: 10.1107/S0567739478001680.

[14] Søren Smidstrup, Andreas Pedersen, Kurt Stokbro, and Hannes Jónsson. Improved initial guess for minimum energy path calculations. *The Journal of Chemical Physics*, 140(21):214106, 2014. doi: 10.1063/1.4878664.

[15] Graeme Henkelman and Hannes Jónsson. Improved tangent estimate in the nudged elastic band method for finding minimum energy paths and saddle points. *The Journal of Chemical Physics*, 113(22):9978–9985, 2000. doi: 10.1063/1.1323224.

[16] Ask Hjorth Larsen, Jens Jørgen Mortensen, Jakob Blomqvist, Ivano E. Castelli, Rune Christensen, Marcin Dułak, Jesper Friis, Michael N. Groves, Bjørk Hammer, Cory Hargus, et al. The atomic simulation environment—a Python library for working with atoms. *Journal of Physics: Condensed Matter*, 29(27):273002, 2017. doi: 10.1088/1361-648X/aa680e.

[17] Diederik P. Kingma and Jimmy Ba. Adam: A method for stochastic optimization. In *International Conference on Learning Representations*, 2015. URL `https://arxiv.org/abs/1412.6980`.

[18] Cloudra. Cloudra source code, 2026. URL `https://github.com/yyywaa/Cloudra-mechanism-agent/tree/70daa5a`. Revision 70daa5a, 26 October 2026.

[19] Charles R. Harris, K. Jarrod Millman, Stéfan J. van der Walt, Ralf Gommers, Pauli Virtanen, David Cournapeau, Eric Wieser, Julian Taylor, Sebastian Berg, Nathaniel J. Smith, Robert Kern, Matti Picus, Stephan Hoyer, Marten H. van Kerkwijk, Matthew Brett, Allan Haldane, Jaime Fernández del Río, Mark Wiebe, Pearu Peterson, Pierre Gérard-Marchant, Kevin Sheppard, Tyler Reddy, Warren Weckesser, Hameer Abbasi, Christoph Gohlke, and Travis E. Oliphant. Array programming with NumPy. *Nature*, 585(7825):357–362, 2020. doi: 10.1038/s41586-020-2649-2.

[20] Noel M. O'Boyle, Adam L. Tenderholt, and Karol M. Langner. cclib: A library for package-independent computational chemistry algorithms. *Journal of Computational Chemistry*, 29(5):839–845, 2008. doi: 10.1002/jcc.20823.

[21] Aric A. Hagberg, Daniel A. Schult, and Pieter J. Swart. Exploring network structure, dynamics, and function using NetworkX. In Gaël Varoquaux, Travis Vaught, and Jarrod Millman, editors, *Proceedings of the 7th Python in Science Conference*, pages 11–15, Pasadena, CA, 2008. URL https://aric.hagberg.org/papers/hagberg-2008-exploring.pdf.

[22] Philipp Pracht, Fabian Bohle, and Stefan Grimme. Automated exploration of the low-energy chemical space with fast quantum chemical methods. *Physical Chemistry Chemical Physics*, 22(14):7169–7192, 2020. doi: 10.1039/C9CP06869D.

[23] Christoph Bannwarth, Sebastian Ehlert, and Stefan Grimme. GFN2-xTB—an accurate and broadly parametrized self-consistent tight-binding quantum chemical method with multipole electrostatics and density-dependent dispersion contributions. *Journal of Chemical Theory and Computation*, 15(3):1652–1671, 2019. doi: 10.1021/acs.jctc.8b01176.

[24] James J. P. Stewart. Optimization of parameters for semiempirical methods VI: more modifications to the NDDO approximations and re-optimization of parameters. *Journal of Molecular Modeling*, 19 (1):1–32, 2013. doi: 10.1007/s00894-012-1667-x.

[25] Axel D. Becke. Density-functional thermochemistry. III. the role of exact exchange. *The Journal of Chemical Physics*, 98(7):5648–5652, 1993. doi: 10.1063/1.464913.

[26] Chengteh Lee, Weitao Yang, and Robert G. Parr. Development of the Colle–Salvetti correlation-energy formula into a functional of the electron density. *Physical Review B*, 37(2):785–789, 1988. doi: 10.1103/PhysRevB.37.785.

[27] P. J. Stephens, F. J. Devlin, C. F. Chabalowski, and M. J. Frisch. Ab initio calculation of vibrational absorption and circular dichroism spectra using density functional force fields. *The Journal of Physical Chemistry*, 98(45):11623–11627, 1994. doi: 10.1021/j100096a001.

[28] Stefan Grimme, Jens Antony, Stephan Ehrlich, and Helge Krieg. A consistent and accurate ab initio parametrization of density functional dispersion correction (DFT-D) for the 94 elements H–Pu. *The Journal of Chemical Physics*, 132(15):154104, 2010. doi: 10.1063/1.3382344.

[29] Stefan Grimme, Stephan Ehrlich, and Lars Goerigk. Effect of the damping function in dispersion corrected density functional theory. *Journal of Computational Chemistry*, 32(7):1456–1465, 2011. doi: 10.1002/jcc.21759.

[30] Florian Weigend and Reinhart Ahlrichs. Balanced basis sets of split valence, triple zeta valence and quadruple zeta valence quality for H to Rn: Design and assessment of accuracy. *Physical Chemistry Chemical Physics*, 7:3297–3305, 2005. doi: 10.1039/B508541A.

[31] Frank Neese. Software update: The ORCA program system—version 6.0. *WIREs Computational Molecular Science*, 15(2):e70019, 2025. doi: 10.1002/wcms.70019.

[32] M. J. Frisch, G. W. Trucks, H. B. Schlegel, G. E. Scuseria, M. A. Robb, J. R. Cheeseman, G. Scalmani, V. Barone, G. A. Petersson, H. Nakatsuji, et al. Gaussian 16. Gaussian, Inc., Wallingford, CT, 2016. URL https://gaussian.com/citation/.

[33] Jonathan E. Moussa and James J. P. Stewart. MOPAC: An open-source semiempirical molecular orbital program. *Journal of Open Source Software*, 11(119):8025, 2026. doi: 10.21105/joss.08025.

[34] FACCTs. ORCA 6.1 Manual, 2025. URL https://www.faccts.de/docs/orca/6.1/manual/. Online documentation; accessed 31 October 2026.

[35] F. Pedregosa, G. Varoquaux, A. Gramfort, V. Michel, B. Thirion, O. Grisel, M. Blondel, P. Prettenhofer, R. Weiss, V. Dubourg, J. Vanderplas, A. Passos, D. Cournapeau, M. Brucher, M. Perrot, and E. Duchesnay. Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research*, 12:2825–2830, 2011. URL https://jmlr.org/papers/v12/pedregosa11a.html.

[36] Jianlyu Chen, Shitao Xiao, Peitian Zhang, Kun Luo, Defu Lian, and Zheng Liu. M3-Embedding: Multi-linguality, multi-functionality, multi-granularity text embeddings through self-knowledge distillation. In *Findings of the Association for Computational Linguistics: ACL 2024*, pages 2318–2335, 2024. doi: 10.18653/v1/2024.findings-acl.137. URL https://aclanthology.org/2024.findings-acl.137/.

[37] DeepSeek-AI. DeepSeek V4 Model Card, 2026. URL https://fe-static.deepseek.com/chat/transparency/deepseek-V4-model-card-EN.pdf.

[38] Luyu Gao, Aman Madaan, Shuyan Zhou, Uri Alon, Pengfei Liu, Yiming Yang, Jamie Callan, and Graham Neubig. PAL: Program-aided language models. In *Proceedings of the 40th International Conference on Machine Learning*, volume 202, pages 10764–10799, 2023. URL https://proceedings.mlr.press/v202/gao23f.html.

[39] Fred Zhang and Neel Nanda. Towards Best Practices of Activation Patching in Language Models: Metrics and Methods. In *International Conference on Learning Representations*, 2024. URL https://proceedings.iclr.cc/paper_files/paper/2024/hash/06a52a54c8ee03cd86771136bc91eb1f-Abstract-Conference.html.

[40] Bahar Fatemi, Bryan Perozzi, and Jonathan Halcrow. Talk like a Graph: Encoding Graphs for Large Language Models. In *International Conference on Learning Representations*, 2024. URL https://research.google/pubs/talk-like-a-graph-encoding-graphs-for-large-language-models/.

[41] Zhibin Gou, Zhihong Shao, Yeyun Gong, Yelong Shen, Yujiu Yang, Nan Duan, and Weizhu Chen. CRITIC: Large language models can self-correct with tool-interactive critiquing. In *International Conference on Learning Representations*, 2024. URL https://arxiv.org/abs/2305.11738.

[42] Karthik Valmeekam, Matthew Marquez, Sarath Sreedharan, and Subbarao Kambhampati. On the planning abilities of large language models: A critical investigation. In *Advances in Neural Information Processing Systems*, 2023. URL https://arxiv.org/abs/2305.15771.

[43] Miles Turpin, Julian Michael, Ethan Perez, and Samuel R. Bowman. Language models don't always say what they think: Unfaithful explanations in chain-of-thought prompting. In *Advances in Neural Information Processing Systems*, 2023. URL https://arxiv.org/abs/2305.04388.

[44] Jie Huang, Xinyun Chen, Swaroop Mishra, Huaixiu Steven Zheng, Adams Wei Yu, Xinying Song, and Denny Zhou. Large language models cannot self-correct reasoning yet. In *International Conference on Learning Representations*, 2024. URL https://arxiv.org/abs/2310.01798.